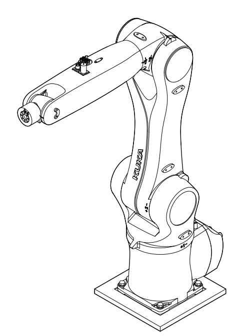
KUKA KR10-1100 ROBOT ARM
MANUAL

The Creative School

Design + Technology LAB



PART 1: FUNDAMENTALS

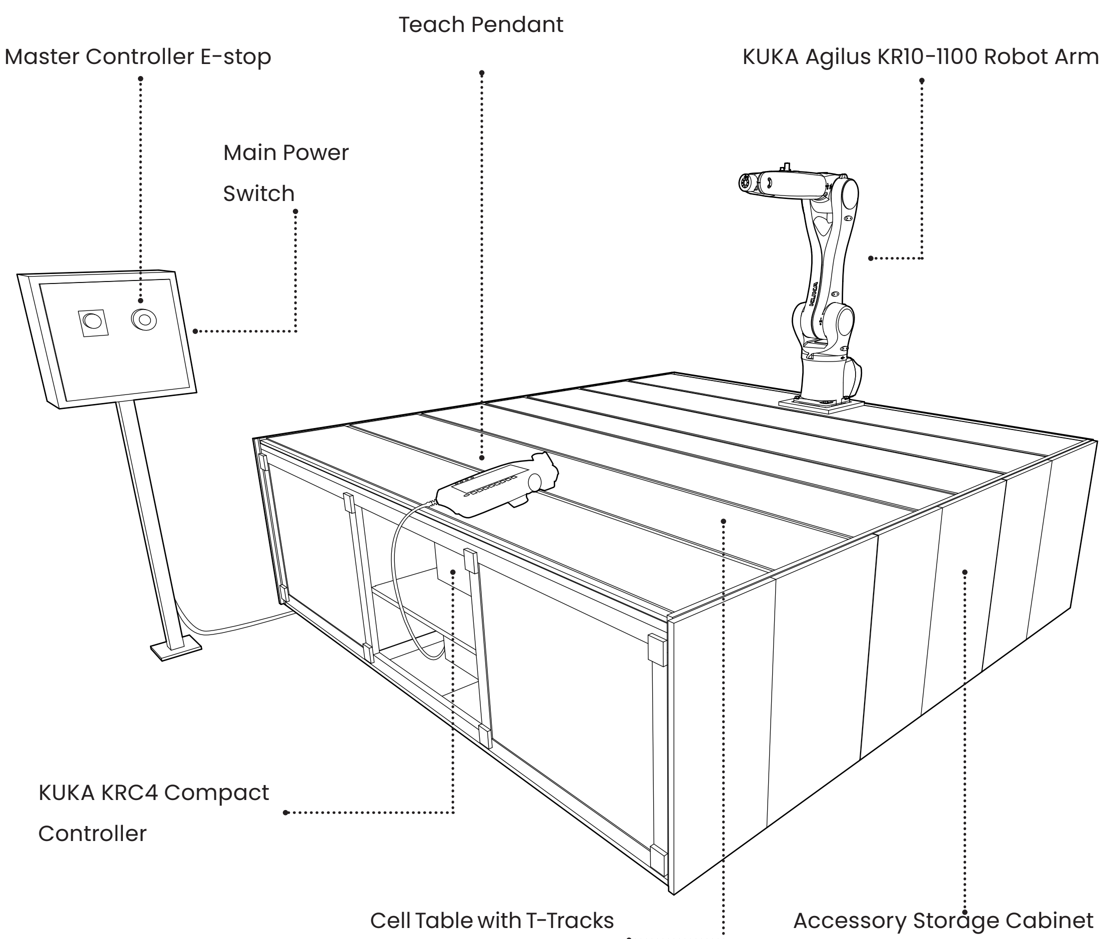
01: The KR10-R1100 Robot Arm

The following manual presents foundational information on the KUKA KR10-R110 Agilus robotic arms at the Design + Technology LAB.

There are two units at the Design + Technology LAB: one located in the Work Room robot cell and the other mounted onto a mobile cell.

Please review this manual and ask a technician to demonstrate proper cutting procedures, before using the machine.

Questions and Concerns: If you have any trouble or questions regarding machine use please email the technicians at: Designtechnologylab@ryerson.ca



Robot cell located in the Work Room

D+TL

General Workshop Safety

Required Personal Protective Equipment (PPE)

Recommended Personal Protective

Emergency Pull Station

Workshop Safety Practices

1. No person will work alone in any areas of the machine room. Design + Technology LAB members working after hours alone must get permission from the lead technician and notify Ryerson security.
2. Safety glasses must be worn while operating any power machinery/tools.
3. Distracting conversations and use of electronics (including cellphones and earbuds) can be dangerous in work areas and must not take place. Please take phone calls in the hall outside the lab.

Clothing and PPE

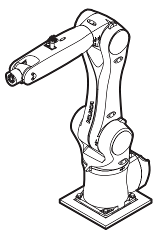
Before entering the Design + Technology LAB: Please ensure that you have read and adhere to the workshop general safety guidelines and are wearing all required PPE. The following rules concern appropriate Workshop apparel:

- Long sleeves must be rolled up.
- Loose scarves or clothing must be secured
- Heavy or loose jewelry must be removed (i.e., rings, watches, bracelets, necklaces, etc.)
- Long hair must be tied back.
- Non-slip, hard-toe, flat shoes must be worn.

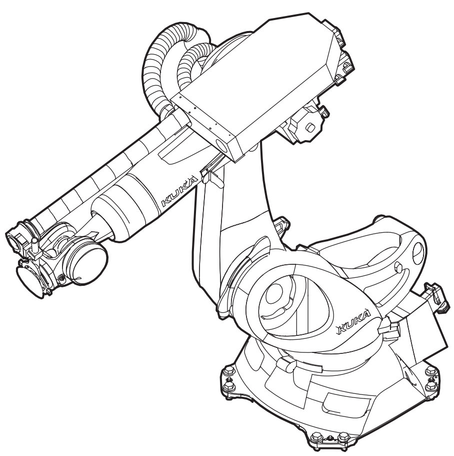
What to do in the case of an accident.

In the case of an accident notify a technician immediately and pull the EPS (Emergency Pull Station) switch to notify security. If possible have the injured person sit down to prevent fainting, collapsing, or further injury.

D+TL
General Workshop Safety



KUKA KR10 Robotic Arm



KUKA KR150 Robotic Arm

Robotic Operation Safety
There are a number of robotic arms located in the Design + Technology LAB. There is 1 large Kuka KR150 in the High Bay (B33A) and 2 Kuka KR10's in the Workroom (B31). Be mindful of robotic operation at the Design + Technology LAB. Do not enter rooms when the following signage is posted.

This sign is most commonly posted on the entrance to the High Bay. Interlocks are installed on doors to the Kuka Cage and High Bay. Do not open doors during Robot use or it will trigger the workshop's Emergency Stop system. If you require access to the Print Lab when a robot in the High Bay is in operation, ask the Design + Technology LAB staff for assistance.

Noise and Air Quality
The Design + Technology LAB is a new facility with a strong interior ventilation system and various dust and fume collectors. Air quality in the facility can be affected by the machining of various materials. The dust and fume collectors work to reduce your risk of exposure to off-gassing materials and fine dusts. In special circumstances, we recommend you wear an N95 mask as an extra precaution.

The dust collectors and fume collectors in the metal and woodworking shop are noisy. Please use earplugs if the noise bothers you. The Design + Technology LAB has disposable ear plugs available.

03:

KUKA

KR10- R1100 Work Cell Safety

Robotics in Design + Technology LAB

The KUKA KR 10 is equipped with an enclosed cell to prevent entry into the active robot work space. The active robot work space is the area within the stopping distances of the manipulator, external axes and end of arm tooling.

The operator has the following responsibilities:

- **Door Signage:** When using the KR150, be sure to post the sign below at BOTH entrances to the high bay. Do not enter rooms when the sign is posted.

- Read the operational procedure guideline at the entrance of the work cell. As well keep the manual in the immediate vicinity of the machine so that it is accessible at all times to all persons working on or with the machine.
- **Interlocks** are installed on the work cell gate. Ensure these are properly closed or it will trigger the emergency stop.
- The machine must not be left unattended while it is operating (supervised operation).
- Switch off the machine at the main switch for periods of non-use. As well turn off the robot integration power that controls the interlocks and safety equipment. You can find this switch at the entrance of the work cell.
- Stop this machine immediately in case of failure.
- In addition to the safety notes and instructions stated in this manual, consider and observe the local accident prevention regulations and general safety regulations that apply at the operation site of the machine.

03:
KUKA
KR10- R1100 Work Cell Safety

Intended use of the industrial robot

The machine described in this manual is intended exclusively for creative activities, teaching, and research. This includes - but is not limited to - 3D printing, plotting, hot wire cutting. Subsequent chapters will reference these techniques using the supplied software and machine accessories.

Machine Information

It is strictly prohibited to alter, refit or modify the machine in any way without the express consent of the manufacturer.

Likewise, it is strictly prohibited to remove the bridge or bypass any safety devices.

Operating conditions and connection and setup values stated in the data sheet must be complied with at all times.

Total mass for load acting **50kg**

Standard Operating Procedure

Step 1: Turn ON the Master control (figure 1), You can find the main switch on the right side of the Master controller.

A. Verify if the robot is ON.

B. When operating using either **T1** or **T2** mode ensure Workspace and Cell area is clear.

03:

KUKA KR10- R1100 Work Cell Safety

Figure 1. Master controller

Step 2: Reset Door (Figure 2), confirm the cell door is closed properly and Safety Interlock lights turn green.

Figure 2. Reset Door

Step 3: To activate the robot: operate your program on either **T1** or **T2** mode to set in motion, once activated turn the key-switch to the right and select **AUTOMATIC** mode.

If robot operation stops due to the following errors, complete the procedure below.

A. Operator safety open - Operator safety has been tripped in the **AUTOMATIC** mode and the cell door is open. This message can be cleared once the door is closed properly, ensuring safety interlocks turn green. Once completed you can press the door reset. To resume your program Switch to mode **T1** or **T2** (in these modes the robot will set in motion) and switch to **AUTOMATIC** mode.

03:

KUKA

KR10- R1100 Work Cell Safety

B. External EMERGENCY STOP - Emergency Stop pushbutton pressed. To resume, Release the Emergency Stop Pushbutton and turn ON the master control inside the cell. Once completed close cell door and press door reset and resume your program. Master control must be turn ON in order to operate using either **T1** or **T2** modes*. To resume your program Switch to mode **T1** or **T2** (in these modes the robot will set in motion) and switch to **AUTOMATIC** mode.

04:

Machine Overview

KR10-R1100 Agilus

KRC4 smartPAD Teach Pendant

There are two KUKA KR10-R110 Agilus robot arm consists of the following components:

The robot are 6-axis jointed-arm manipulators made of cast light alloy. Each axis is fitted with a brake. All motor units and current carrying cables are protected against dirt and moisture beneath screwed-on cover plates.

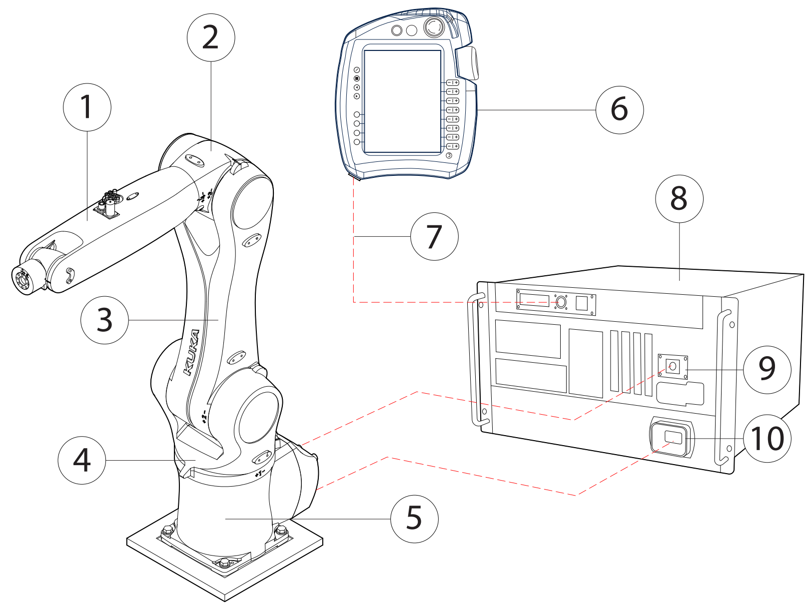


Figure 1: Overview

1. **In-line wrist:** The robot is fitted with a 3-axis in-line wrists. It consists of axes A4, A5 and A6. There are three 5/2-way solenoid valves and a CAT5 data cable in the in-line wrist that can be used for controlling tools. The in-line wrist also accommodates the 10-contact circular connector of the wrist I/O cable and interface A4 for the energy supply system. Refer to *Figure 2: Axis Diagram*.

2. **Arm:** The arm is the link between the in-line wrist and the link arm. The arm is driven by the motor of axis 3.

04:

Machine Overview

KR10-R1100 Agilus

KRC4 smartPAD Teach Pendant

3. **Link arm:** the assembly located between the arm and the rotating column. It houses the motor and gear unit of axis 2. The supply lines of the energy supply system and the cable set for axis 2 to 6 are routed through the link arm.
4. **Rotating column:** houses the motors of axes 1 and 2. The rotational motion of axis 1 is performed by the rotating column. This is screwed to the base frame via the gear unit of axis 1 and is driven by a motor in the rotating column. The link arm is also mounted in the rotating column.
5. **Base frame:** the base of the robot. Interface A1 is located at the rear of the base frame. It constitutes the interface for the connecting cables between the robot, the controller and the energy supply system.
6. **SmartPAD Teach Pendant:** This is the main human interface of the manipulator with which the operator can move and program the robot.
7. **Teach Pendant Connecting Cable:** This cable connects the teach pendant to the KRC4 Compact Controller. It relays information from the pendant to the drivers within controller
8. **KRC4 Compact Controller:** This box contains the drivers for every stepper motor within the robot arm. It functions as the brain of the robot arm, processing code into motor rotations.
9. **Data Cable:** This cable connects the KRC Compact Controller to the robot arm creating a feedback loop between the two systems.
10. **Motor Cable:** This cable connects also connects the KRC Compact Controller to the robot arm. It relays power from the controller to the arm allowing its motors to engage.

04:

Machine Overview

KR10-R1100 Agilus

KRC4 smartPAD Teach Pendant

AXIS OVERVIEW

The KR10-R1100 has six axis, each with varying degrees of motion. Each axis (annotated below) can engage simultaneously to move and rotate an end effector anywhere within its reach. It is important to have an idea of where each axis is located as you program movements in Grasshopper and while operating the machine.

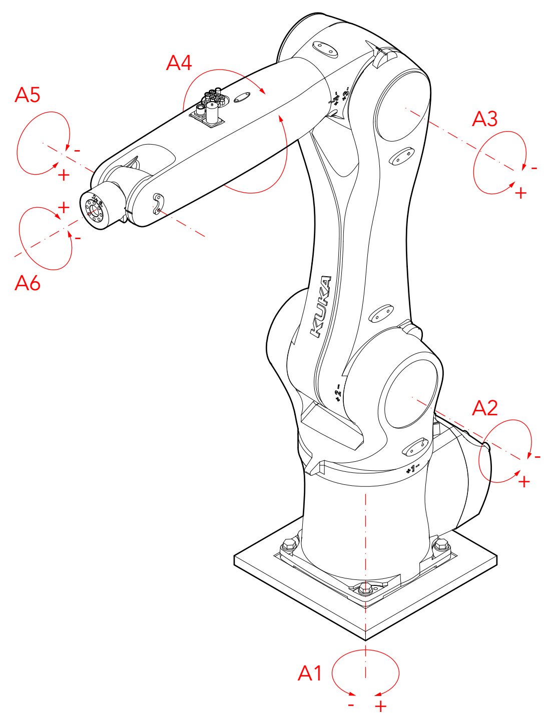


Figure 2: Axis Diagram

04:

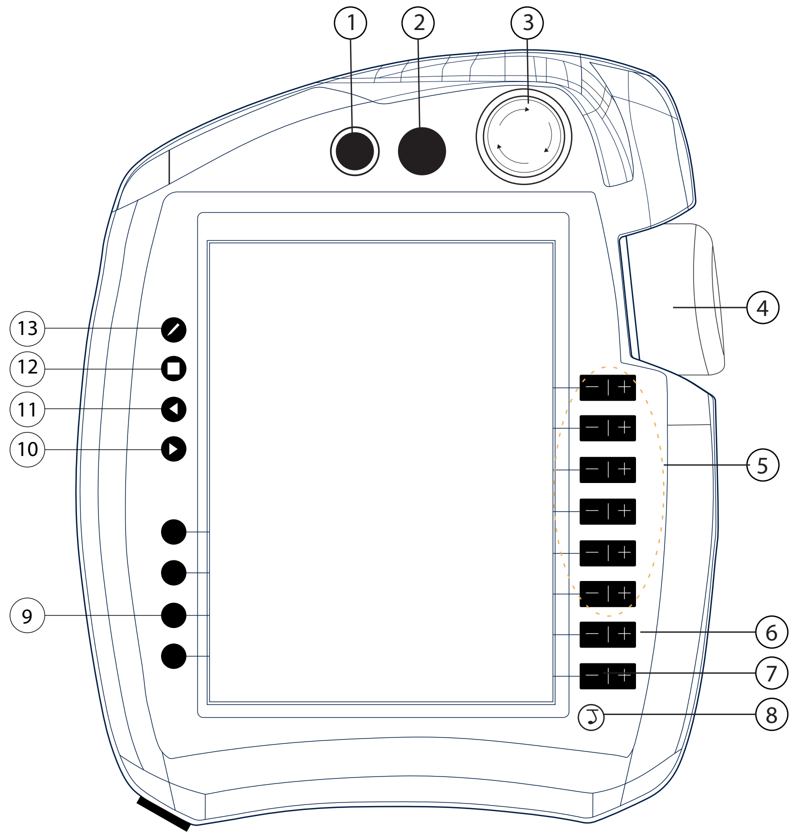
Machine Overview

KR10-1100 Agilus

KRC4 smartPAD Teach Pendant

KUKA KRC4 smartPAD teach pendant

The smartPAD is a teach pendant for the industrial robot. The smartPAD has all the operator controls and display functions required for operating and programming the industrial robot. It is equipped with a touch screen and features listed below.



FRONT VIEW

1. Button for disconnecting the smartPAD
2. **Key switch** for calling the connection manager. This switch can only be turned if the key is inserted. Operating modes can be changed by using the connection manager.
3. **EMERGENCY STOP** button stops the robot in hazardous situations. The EMERGENCY STOP

04:

Machine Overview

KR10-1100 Agilus

KRC4 smartPAD Teach Pendant

button locks itself in place when it is pressed.

4. **3D Mouse:** For moving the robot manually.
5. **Jog keys:** For moving the robot manually.
6. Key for setting the **program override**
7. Key for setting the **jog override**
8. **Main menu key:** Shows the menu items on the smartPAD
9. **Status keys.** The status keys are used primarily for setting parameters in technology packages. Their exact function depends on the technology packages installed.
10. **Start key:** The Start key is used to start a program.
11. **Start backwards key:** The Start backwards key is used to start a program backwards. The program is executed step by step.
12. **STOP key:** The STOP key is used to stop a program that is running.
13. **Keyboard key** displays the keyboard. It is generally not necessary to press this key to display the keyboard, as the smartPAD detects when keyboard input is required and displays the keyboard automatically.

Note

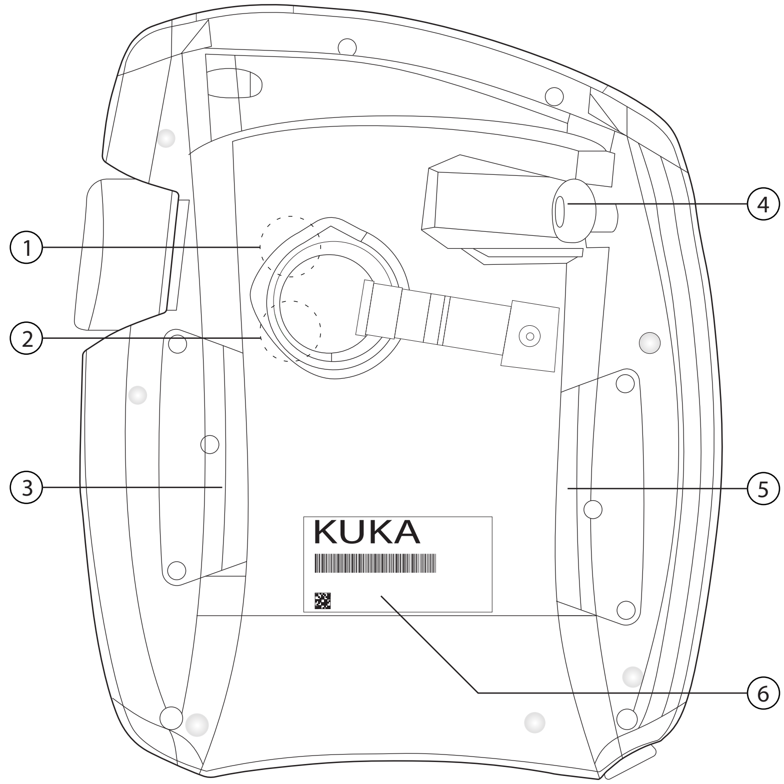
- The enabling switch must be held in the center position in operating modes T1 and T2 in order to be able to jog the manipulator.
- In the operating Automatic mode, the enabling switches have no function.
- The USB connection is used for transferring data.

04:

Machine Overview

KR10-1100 Agilus

KRC4 smartPAD Teach Pendant



REAR VIEW

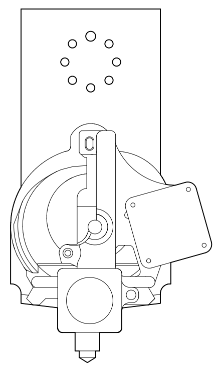
1. Enabling switch
2. Start key (green)
3. Enabling switch
4. USB connection
5. Enabling switch
6. Identification plate

05:

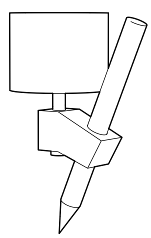
End Effectors

The KR10-R1100 is capable of manipulating an end effector with a payload of up to 50kg (110lbs). The possibilities of what you can mount on the KUKA robotic arm are virtually endless as long as it does not exceed this maximum payload. The following end effectors are available for use at the CTL. Detailed instructions on how to use the following tools will be outlined in *Part 2: Fabrication.*

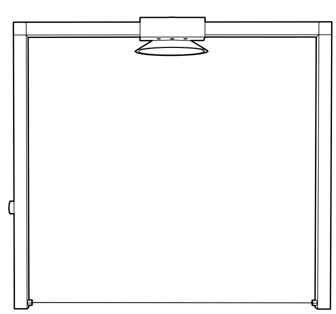
FDM 3D Print Extruder



Pen Plotter



Hot Wire Cutter



05: End Effectors

To change or mount an end effector please consult a Design + Technology LAB Technologist. DO NOT change an end effector that is currently mounted on the robotic arm without the approval of a Design + Technology LAB Technologist.

To create a custom end effector please review your tool design and fabrication process with a Design + Technology LAB Technologist. ALL end effectors must be weighed before being mounted onto the robotic arm. The robotic arm can only accept M10 machine screws so design your mounting plate accordingly.

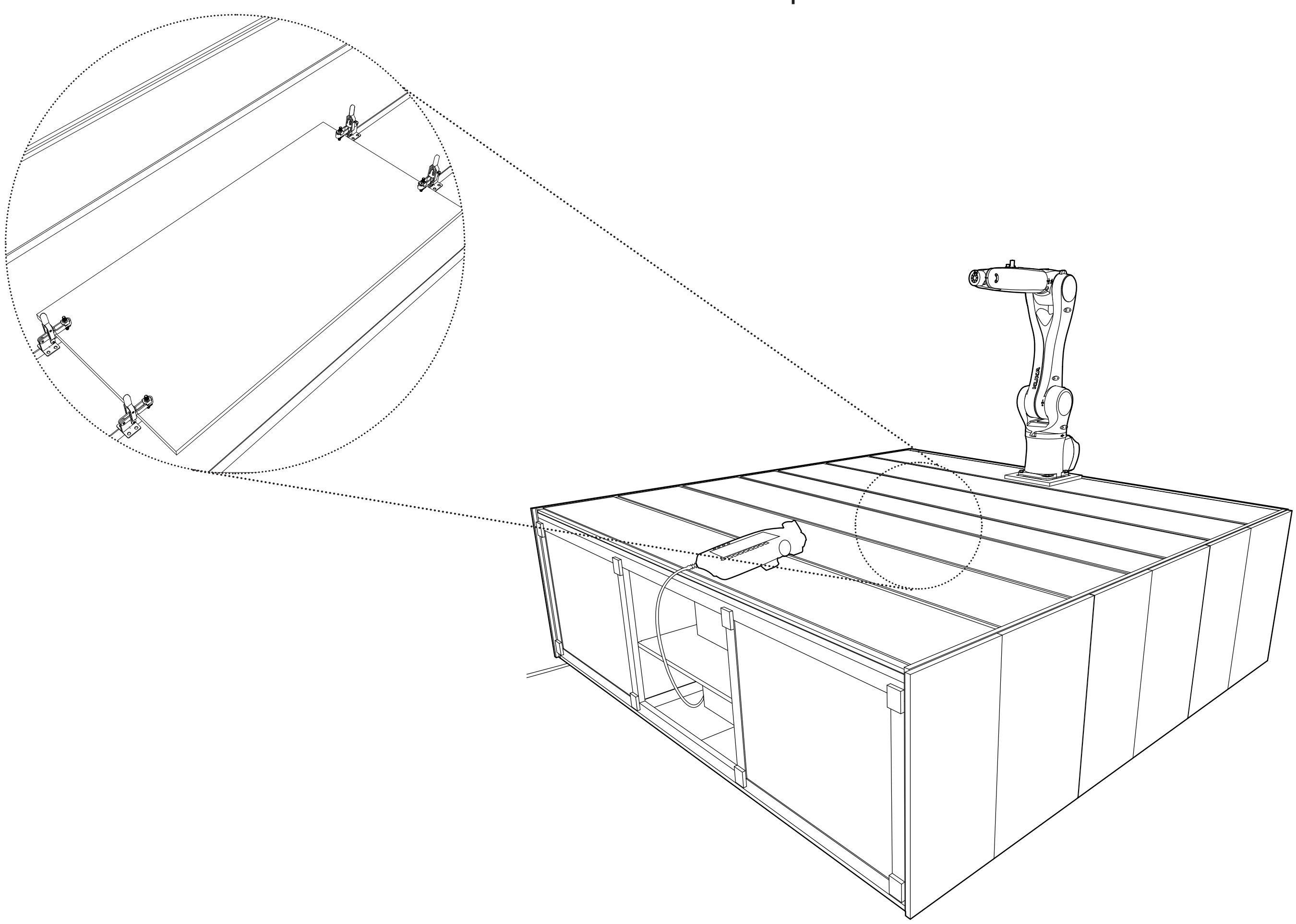
06: Work Holding

Your work piece should be COMPLETELY SECURED using the following work holding techniques before operating the robot.

Please consult a Design + Technology LAB Technologist to ensure that you are using the appropriate work holding techniques.

Cell Table

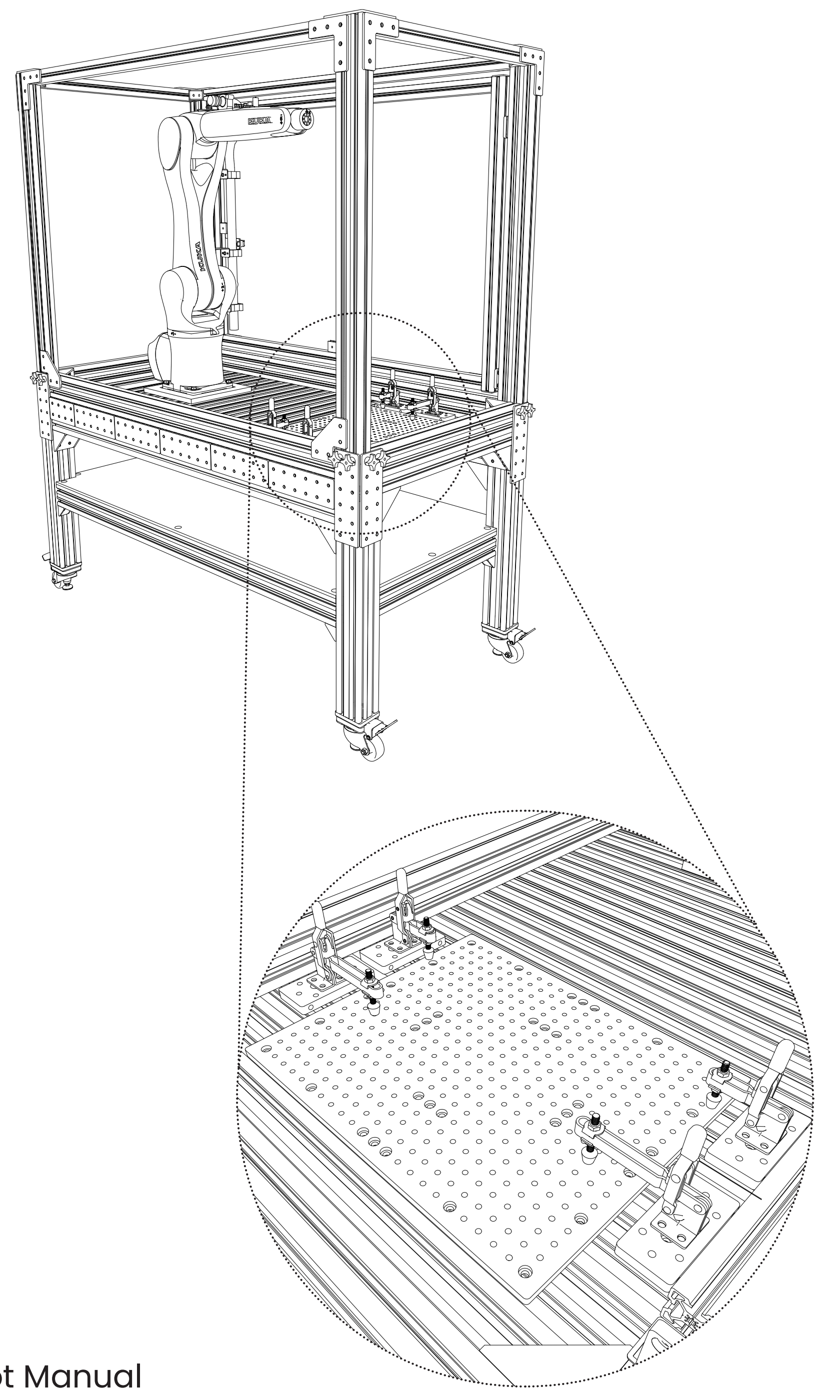
The KR10-R1100 Agilus robots can be secured directly onto the bed of the cell table. Integrated aluminum T-tracks allow you to clamp down and secure a solid board to the bed. Depending on the nature of your project, you may need to design and build a custom mount that is specific to your work piece and can be clamped to the bed.



06: Work Holding

Vention Mobile Cell

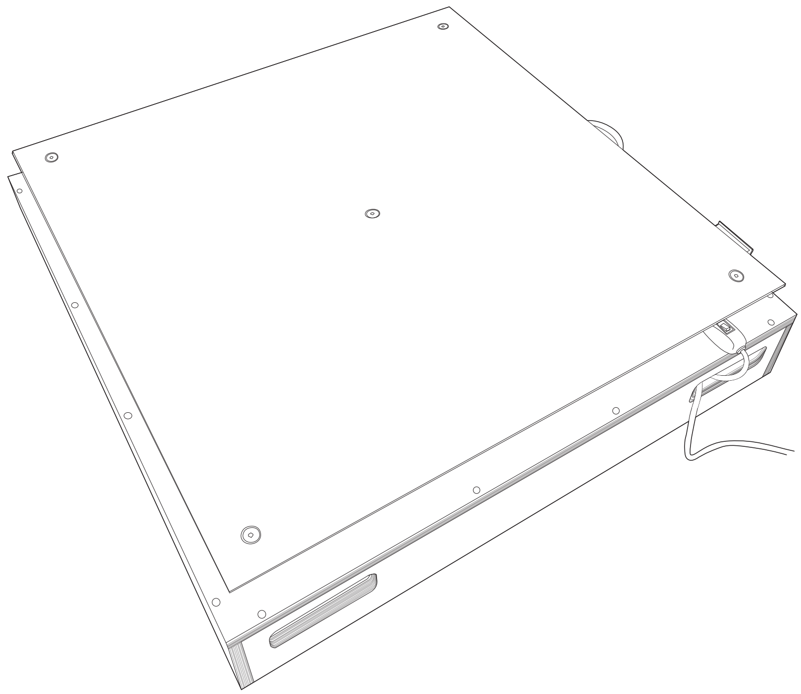
The KR10-R1100 Agilus robots can also be secured to the Vention mobile cell. This unit functions in the same manner as the cell table and has more T-track mounts available. However, it is limited to a smaller working area. It can be moved to any location within the Design + Technology LAB.



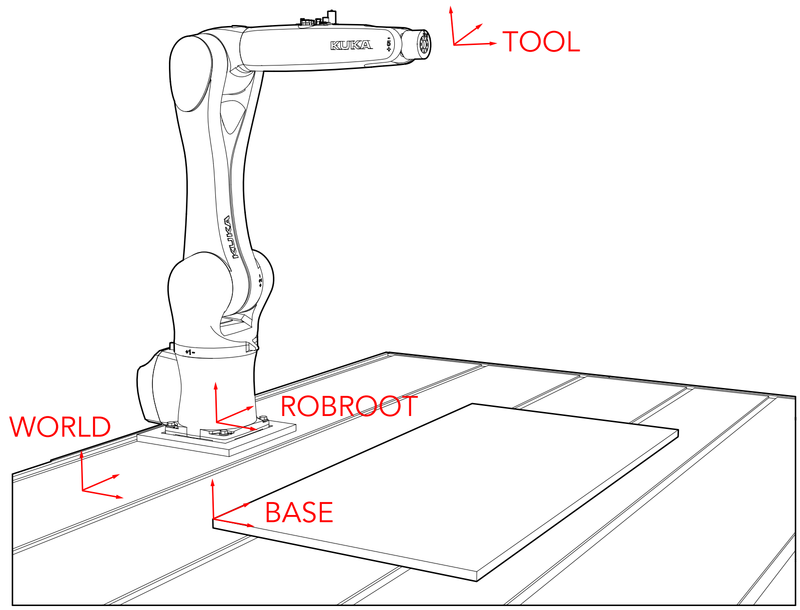
06: Work Holding

3D Printing Hot Plate

The 3D Printing Hot Plate can be used with both the KR10R-1100 and the KR150 robot arm. Be sure that the aluminum plate is leveled accordingly. CAUTION: the surface of the plate can be hot.



07: Coordinate Systems

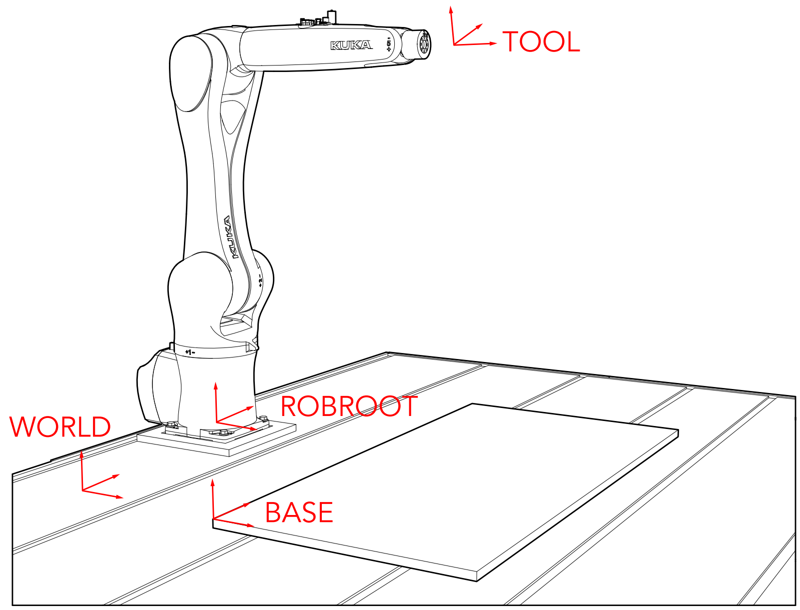


WORLD
The WORLD coordinate system is a permanently defined Cartesian coordinate system. It is the root coordinate system for the ROBROOT and BASE coordinate systems. By default, the WORLD coordinate system is located at the robot base.

ROBROOT
The ROBROOT coordinate system is a Cartesian coordinate system, which is always located at the robot base. It defines the position of the robot relative to the WORLD coordinate system. By default, the ROBROOT coordinate system is identical to the WORLD coordinate system. ROBROOT allows the definition of an offset of the robot relative to the WORLD coordinate system.

BASE
The BASE coordinate system is a Cartesian coordinate system that defines the position of the workpiece. It is relative to the WORLD coordinate system. By default, the BASE coordinate system is identical to the WORLD coordinate system. It is offset relative to the workpiece by the user.

07:
Coordinate
Systems



TOOL
The TOOL coordinate system is a Cartesian coordinate system which is located at the tool center point. It is relative to the BASE coordinate system.

08:

The Robotic Fabrication Process

Your design could be digitally modeled in Rhino with/or without the use of Grasshopper. However, it will be processed through the **KUKA PRC (Parametric Robotic Control) plug-in on Grasshopper** to generate movements to control the robotic arm.

The following is a general overview of the Grasshopper and KUKA PRC interface. It will outline key components for generating a script. Each script will be completely different depending on what fabrication method you are working with. In-depth instructions of each method will be outlined in *Part 2: FABRICATION.*

Grasshopper Interface

Grasshopper may not already have the KUKA PRC plug-in already installed. You may need to register and download the software from:

https://www.robotsinarchitecture.org/kukaprc

Please speak to a Design + Technology LAB Technologist to acquire the proper license to be used with the software.

PRC stands for Parametric Robot Control.
KUKA PRC is the primary robotic programming software used for all KUKA robotic arms at the CTL. This plug-in has 5 main components in the Grasshopper interface, each with its own set of functions and commands:

1. Core
2. Virtual Robot
3. Virtual Tool
4. Toolpath Utilities
5. Utilities

08:

The Robotic Fabrication Process

STEP 1: KUKA PRC | Grasshopper Programming

STEP 2: KRC4 smartPAD Teach Pendant

STEP 3: Operating Mode Selection

STEP 4: Tool Calibration

STEP 5: Import and Run a Program

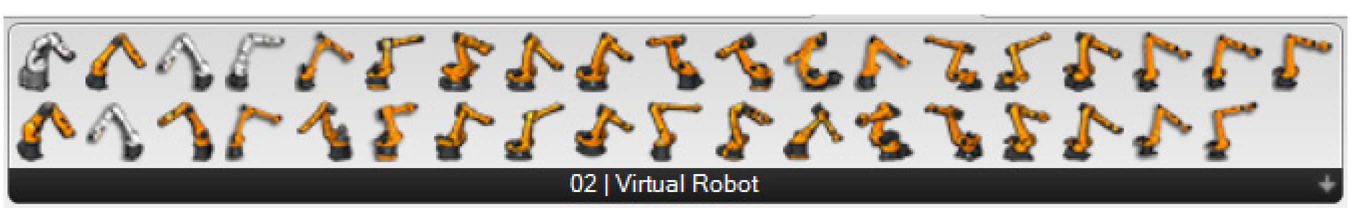
STEP 6: Powering off the Machine

01 | Core

KUKA Core provides fundamental controls and movements for the virtual robot. The component is the central hub through which your Grasshopper script is processed (this will be explained *further in KUKA PRC Interface*). Advanced users may also code KUKA Robotic Language (KRL) directly into Grasshopper with the component.

There are five basic robotic movements available:

1. **6-Axis (6A):** movement defined by axis rotation values
2. **Linear (LIN):** movement through a plane in a straight line
3. **Spline (SPL):** robot path interpolated as a spline through a list of planes.
4. **Circular (CIR):** movement defined through an arc
5. **Point-to-point (PTP):** movement defined between two points

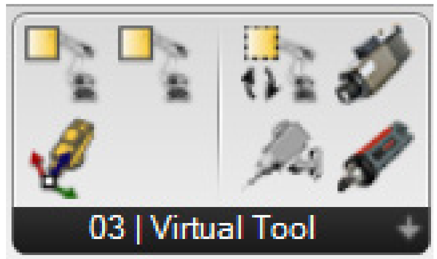


02 | Virtual Robot

This menu provides a variety of robot models to choose from. You may drag and drop the robot you are working with into the script from here.

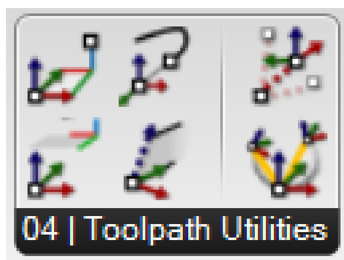
08:

The Robotic Fabrication Process



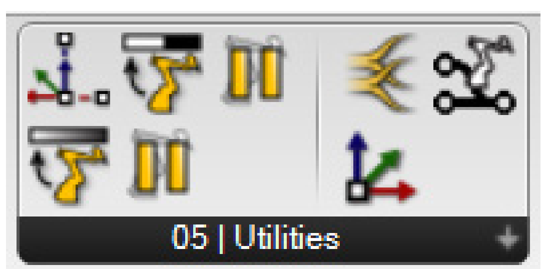
03 | Virtual Tool

This menu is where you can equip the virtual robot with a specific end effector. There are several default tools as well as an option to create a custom tool. A Design + Technology LAB Technologist can provide you with the corresponding tool data for the end effector you are working with.



04 | Toolpath Utilities

This menu provides additional movement commands. These components offer a higher level of complexity than the fundamental movements.



05 | Utilities

This menu provides additional components that can help manage the data that is being processed in your script.

08:

The Robotic Fabrication Process

STEP 1: KUKA PRC | Grasshopper Programming

STEP 2: KRC4 smartPAD Teach Pendant

STEP 3: Operating Mode Selection

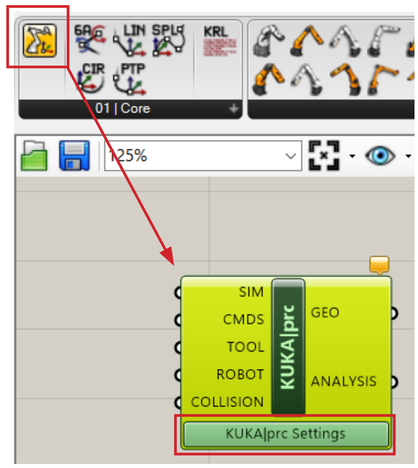
STEP 4: Tool Calibration

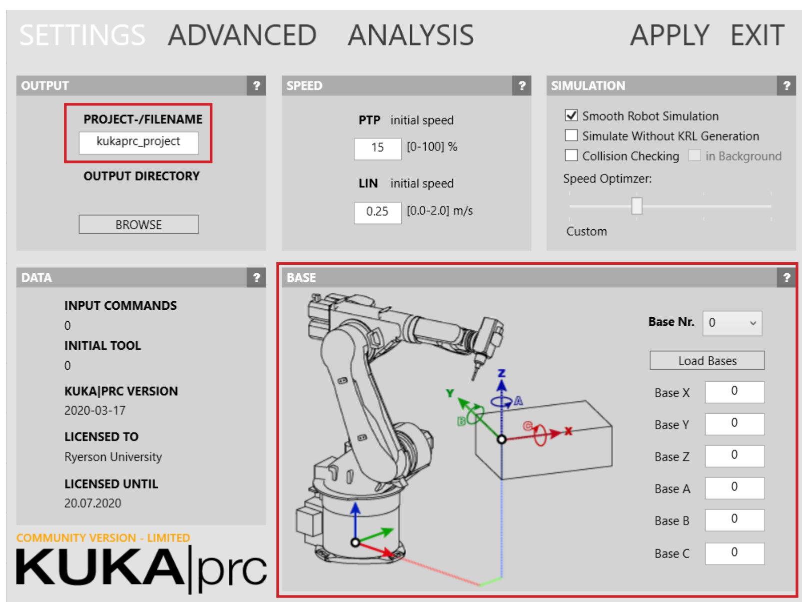
STEP 5: Import and Run a Program

STEP 6: Powering off the Machine

KUKA PRC Interface

To access the KUKA PRC hub, drag and drop the component from *01 | Core* into your script. Click *KUKA|PRC Settings* to access the hub window. The following menus will highlight the most important areas.



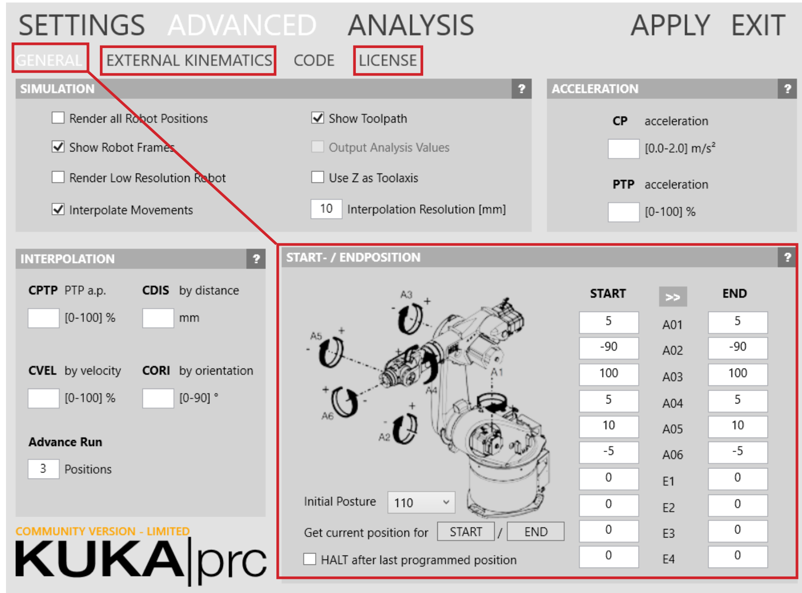


Settings Menu

- **Project-/File name:** this is the file location where your code will be saved
- **Base:** This is where you will input base coordinates.

08:

The Robotic Fabrication Process



Advanced Menu

- **General > Start/End Position**: This is where you will input information to position and orient a tool onto the virtual robot
- **External Kinematics:** only use this menu when working with a turntable or rail
- **License:** this is where you will input a provided license file

08:

The Robotic Fabrication Process

STEP 1: KUKA PRC | Grasshopper Programming

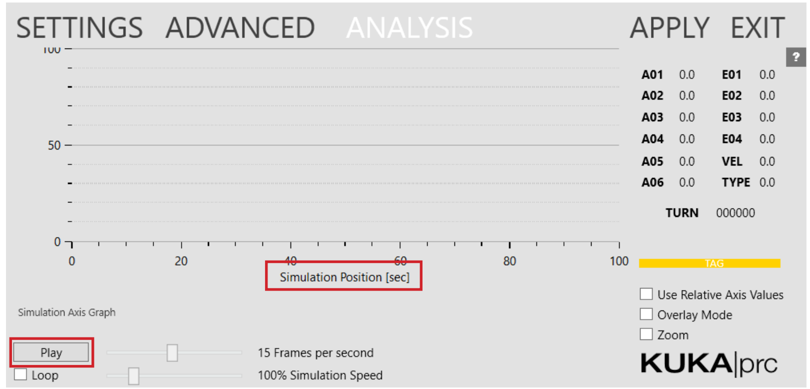
STEP 2: KRC4 smartPAD Teach Pendant

STEP 3: Operating Mode Selection

STEP 4: Tool Calibration

STEP 5: Import and Run a Program

STEP 6: Powering off the Machine



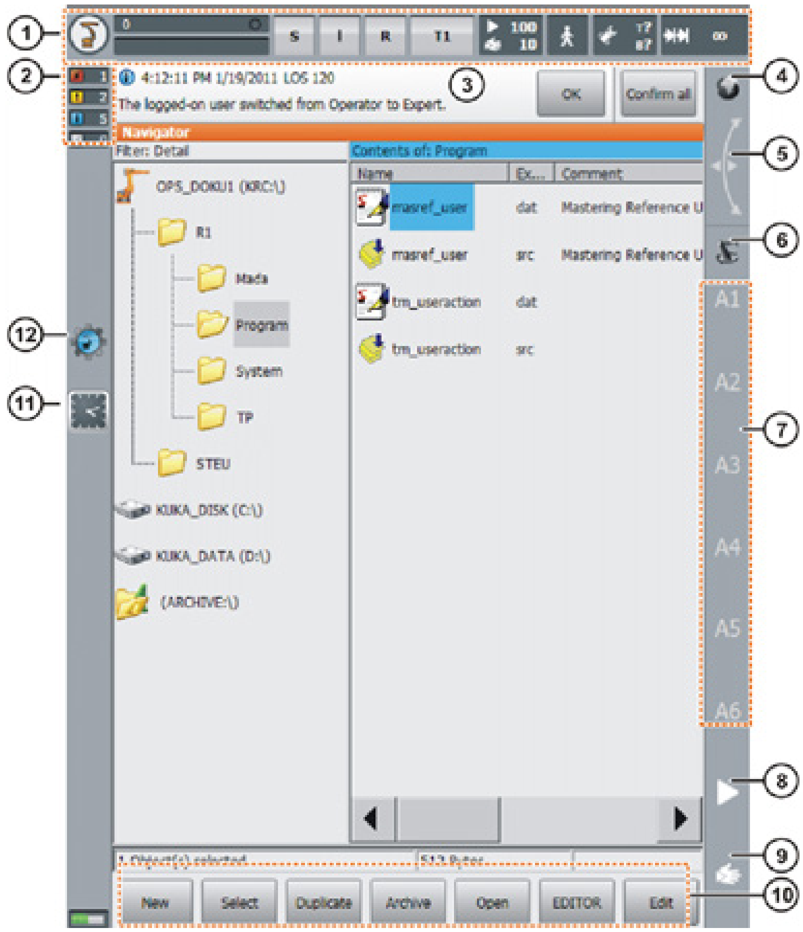
Analysis Menu

- Press **Play** to simulate the virtual robot. The performance of your script will be displayed in real time on the *Simulation-Position graph.* Areas indicating collisions will appear in red.

08:

The Robotic Fabrication Process

KRC4 Compact Controller Interface



1. **Status bar**
2. **Message counter:** this indicates how many messages of each message type are active. Touching the message counter expands the messages.
3. **Message window:** by default, only the last message is displayed. Touching the message window expands it so that all active messages are displayed. An acknowledgment message can be acknowledged with OK. All knowledge-able messages can be acknowledged at once with All OK.
4. **Space mouse:** this indicator shows the current coordinate system while jogging the robot with the Space mouse. Touching

08:

The Robotic Fabrication Process

the indicator displays all coordinate systems, allowing a different one to be selected.

5. **Space mouse alignment:** touching this indicator opens a window that displays the current alignment of the Space Mouse. Alignment can be changed here as well.

6. **Jog keys:** this indicator shows the current coordinate system while jogging the robot with jog keys. Touching the indicator displays all co-ordinate systems, allowing a different one to be selected.
   i) Axes mode limits the Kuka movements to the six axes.

   ii) World mode restricts the Kuka movement to the X, Y, Z axes, calculating the complex kinematics (of all six axes in order to do so).

7. **Jog key labels:** If axis-specific jogging is selected, the axis numbers are displayed here (A1, A2, etc.). If Cartesian jogging is selected, the coordinate system axes are displayed here (X, Y, Z, A, B, C).

8. **Program override:** the override can be set using the plus/minus key on the right hand side of the screen. The value can be set to 100%, 75%, 50%, 30%, 10%, 3%, 1%.

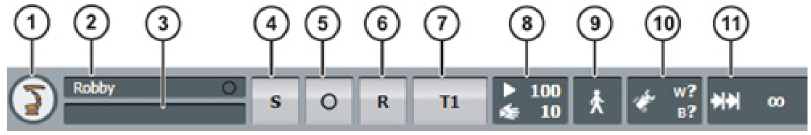
9. **Jog override:** Jog override determines the velocity of the robot while jogging. It allows you to change the speed of the machine movements before the program sequence begins. Plus/minus keys: The value can be set to 100%, 75%, 50%, 30%, 10%, 3%, 1%

10. **Clock:** the clock displays the system time.

11. **Work Visual icon:** if no project can be opened: the icon has a small red X in the bottom left-hand corner.

08:

The Robotic Fabrication Process



Status bar

1. **Main menu key:** Shows the menu items on the smartPAD.

2. **Robot name:** If a program has been selected, the name is displayed here.

3. **Submit interpreter status indicator** (Figure. 1)

Icon	Color	Description
S	Yellow	Submit interpreter is selected. The block pointer is situated on the first line of the selected SUB program.
S	Green	Submit interpreter is running.
S	Red	Submit interpreter has been stopped.
S	Gray	Submit interpreter is deselected.

Figure. 1

4. **Drives status indicator:** The drives can be switched on or off here.

5. **Robot interpreter status indicator:** Programs can be reset or canceled here. (Figure. 2)

Icon	Color	Description
R	Gray	No program is selected.
R	Yellow	The block pointer is situated on the first line of the selected program.
R	Green	The program is selected and is being executed.
R	Red	The selected and started program has been stopped.
R	Black	The block pointer is situated at the end of the selected program.

Figure. 2

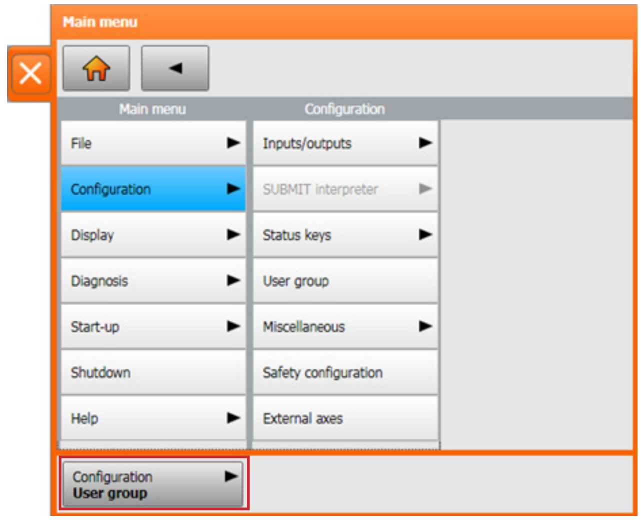
08:

The Robotic Fabrication Process

6. Current operating mode
7. POV/HOV status indicator. Indicates the current program override and the current jog override.
8. Program run mode status indicator. Indicates the current program run mode.
9. Tool/base status indicator. Indicates the current tool and base.
10. Incremental jogging status indicator.

08:

The Robotic Fabrication Process



User group

Different functions are available depending on the user group. The following user groups are available:

In the main menu select Configuration > User group. The current user group is displayed.

Operator: User group for the operator. This is the default user group.

User: User group for the operator. (By default, the user groups “Operator” and “User” are defined for the same target group.)

Expert: User group for the programmer. This user group is protected by means of a password. By changing to Expert mode you can import files from a USB.

Changing user group

Select the menu sequence Configure > User group. The current user group is displayed.

Select the new user group by pressing the corresponding soft key.

08:

The Robotic Fabrication Process

If prompted: Enter password and confirm with OK. Ask a Design + Technology LAB Technologist for the password.

When the system is booted, the user group "User" is selected by default. The user groups "Expert" and "Administrator" are password-protected. Press Default to switch to the default user group; however, it is not available if the default user group is already selected. Please refer to figure. 1:

Action	Possible in user group ...?
Insert comment or stamp	**User**: Yes **Expert**: Yes
Delete lines	**User**: Yes **Expert**: Yes
Create folds	**User**: No **Expert**: Yes
Copy	**User**: No **Expert**: Yes
Paste	**User**: No **Expert**: Yes
Insert blank lines (press the Enter key)	**User**: No **Expert**: Yes
Cut	**User**: No **Expert**: Yes
Find	**User**: Yes **Expert**: Yes Possible for all user groups in an open program, even in AUT EXT mode.
Replace	**User**: No **Expert**: Yes (program is open, not selected)
Programming with inline forms	**User**: Yes **Expert**: Yes
KRL programming	**User**: Possible to a certain extent. KRL instructions covering several lines (e.g. LOOP ... ENDLOOP) are not permissible. **Expert**: Yes

Figure. 1

08:

The Robotic Fabrication Process

Operating modes

Key switch for calling the connection manager

In Manual Reduced Velocity mode (T1):

- If it can be avoided, there must be no other persons inside the safeguarded area.
- If it is necessary for there to be several persons inside the safeguarded area, the following must be observed:
- Each person must have an enabling device.
- All persons must have an unimpeded view of the industrial robot.
- The operator must be so positioned that they can see into the danger area and get out of harm's way.

In Manual High Velocity mode (T2):

Manual mode may only be used if the following conditions are met:

- Application requires a test at a velocity higher than Manual Reduced Velocity.
- Teaching and programming are not permissible in this operating mode.
- Before commencing the test, the operator must ensure that the enabling devices are operational.
- The operator must be positioned outside the danger zone (ie. outside the working cell).
- It is the responsibility of the operator to ensure that there are no other persons inside the safeguarded area.

08:

The Robotic Fabrication Process

Automatic mode – key enabled only

Automatic mode is only permissible in compliance with the following safety measures:

- All safety equipment and safeguards are present and operational.
- The key must be obtained from a tech and must be returned after use
- No personnel should be inside the work cell.
- If the manipulator or an external axis (optional) comes to a standstill for no apparent reason, you should not enter the working cell until an EMERGENCY STOP has been triggered.
- Robot must NEVER be left in automatic mode after intended use.

Overview of operating modes and safety functions

The following table indicates the operating modes in which the safety functions are active.

Safety functions	T1, CRR	T2	AUT	AUT EXT
Operator safety	-	-	active	active
EMERGENCY STOP device	active	active	active	active
Enabling device	active	active	-	-
Reduced velocity during program verification	active	-	-	-
Jog mode	active	active	-	-
Software limit switches	active	active	active	active

08:

The Robotic Fabrication Process

TCP Tool calibration: XYZ 4-Point Procedure

Calibrations must be done in operating mode **T1** or **T2**.

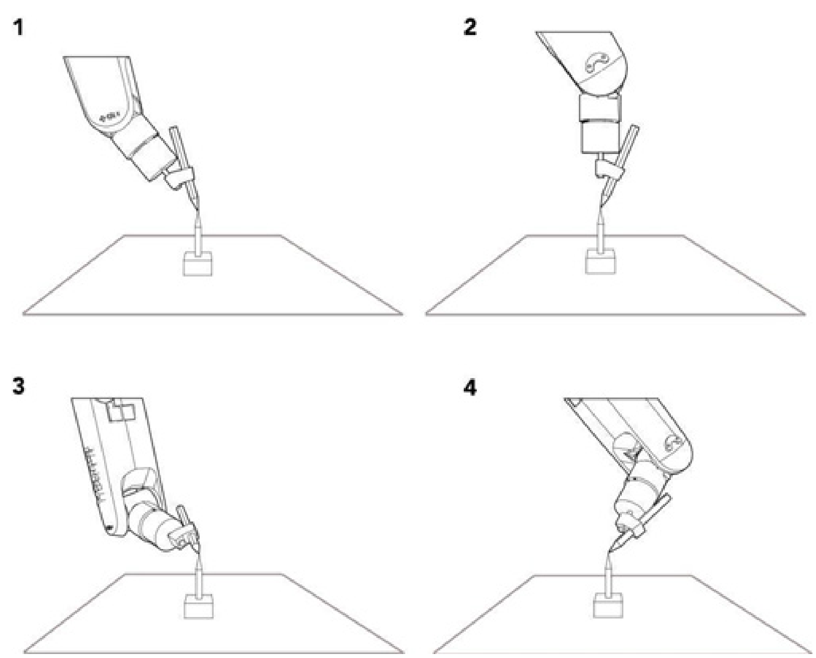
1. Select Start-up > Calibrate > Tool >XYZ 4 points.
2. Assign a number and a name for the tool to be calibrated. Confirm with OK.
3. Move the TCP to the reference point tool shown below. Confirm with OK.
4. Move the TCP to the reference point from a different a direction and confirm with OK. This must be done a total of four times. Always complete XYZ 4 - Point in World mode.

    1) X, Y, Z = movement around these axes

    2) A = Rotation around the X-axis

    3) B = Rotation around the Y-axis

    4) C = Rotation around the Z-axis



08:

The Robotic Fabrication Process

5. Press Save.
6. Press Calibrate, then move the robot away, change its orientation and move it to the same point from another direction, repeat this a total of 4 times. To get actuate results change Incremental Jogging Mode from Continuous to shorter distance. Setting are:

- 100 mm
- 10 mm
- 1 mm
- 0.1 mm

7. Upon completion, the robot asks you to enter the weight of the tool, make sure to insert it in Kg. Make a note of the X, Y and Z numbers. The average error of your measurement should be 0.5 or below to have an accurate tool.
8. Open up KUKA PRC grasshopper and enter the previous XYZ 4 point values under the end effector Tool component. Make sure to copy the exact same values from the previous step.

Edit Calibrated tool

In the main menu select Start-up > Numeric Input: to access Tool XYZ 4-Point values. Here, you can also edit the weight of the tool.

Import Files

Plug in the USB stick in the back of the teach pendant. In expert mode you will grant access to the USB folder in the directory. Select and copy your .SRC file, then go back to the directory and paste your file in the R1 folder.

Your program must be in the R1 folder in order for the robot to run the program.

Program is selected:

- The block pointer is displayed.
- The program can be started.
- The program can be edited to a certain extent. Selected programs are particularly suitable for editing in the user group “User”. Example: KRL instructions covering several lines (e.g. LOOP … ENDLOOP) are not permissible.
- When the program is deselected, modifications are accepted without a request for confirmation. If impermissible modifications are programmed, an error message is displayed.

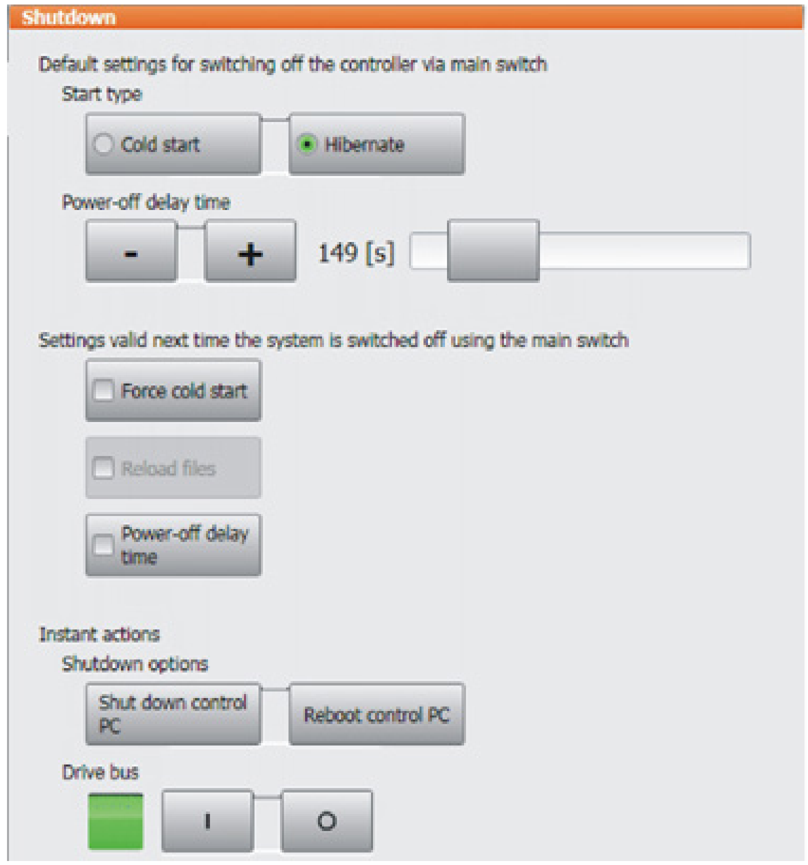
Program is opened:

- The program cannot be started.
- The program can be edited. Opened programs are particularly suitable for editing in the user group “Expert”.
- A request for confirmation is generated when the program is closed. Modifications can be accepted or rejected.

08:

The Robotic Fabrication Process

Shutdown



Select the menu item SHUTDOWN in the main menu.

Select the desired options.

Press **Shutdown control PC** or **Reboot control PC**.

Confirm the request for confirmation with Yes. The System Software is terminated and restarted in accordance with the selected option.

Cold start: after a power failure, the robot controller starts with a cold start.

Hibernate: after a power failure, the robot controller starts with Hibernate.

Power-off delay time: Wait time before the robot controller is shut down. This ensures that the system does not immediately shut down, for example, in the event of very brief power failures, but bridges the power failure.

08:

The Robotic Fabrication Process

Force cold start: When ACTIVATED, the next start is a cold start. This function is only available if *Hibernate* has been selected.

Reload files: The next start is an initial cold start. This option must be selected in the following cases:

- If XML files have been changed directly, i.e. the user has opened the file and modified it. (Any other changes to XML files, e.g. if the robot controller modifies them in the background, are irrelevant.)
- If hardware components are to be exchanged after shutdown. Can only be selected in the *Expert* user group. Only available if *Cold start* or *Force cold* start has been selected.

Power-off delay time: When ACTIVATED, the wait time is applied to the next time the system is shut down. When DEACTIVATED, the wait time is ignored the next time the system is shut down.

Shut down control PC: only available in operating modes T1 and T2. The robot controller is shut down.

Reboot control PC: only available in operating modes T1 and T2. The robot controller is shut down and then restarted.

Drive bus OFF / ON: only available in operating modes T1 and T2. The drive bus can be switched off or on.

Drive bus status indicator:

- Green: Drive bus is on.
- Red: Drive bus is off.
- Gray: Status of the drive bus is unknown.

09: Glossary

Axis range: Range of each axis, in degrees or millimeters, within which it may move. The axis range must be defined for each axis.

Work Cell: All the equipment needed to perform the robotic process (robot, table, fixtures, etc.)

Work Envelope: All the space the robot can reach.

KCP (KUKA Control Panel): Teach pendant KRC4 for KUKA KR10-R1100 and KRC2 for KUKA KR150. The KCP has all the operator control and display functions required for operating and programming the industrial robot.

T1: Test mode, operating Manual Reduced Velocity (<= 250 mm/s)

T2: Test mode, operating Manual High Velocity (> 250 mm/s permissible)

Automatic: operating at full velocity

Degrees of Freedom: The number of movable motions in the robot. To be considered a robot there needs to be a minimum of 4 degrees of freedom. The Kuka Agilus robots have 6 degrees of freedom.

Payload: The amount of weight a robot can handle at full arm extension and moving at full speed.

EOAT - End of Arm Tooling: Tools or Attachments that can be mounted at the end of the robot such as: Pen, 3D extruder, Pellet extruder, Welding gun, gripper, etc. Also known as an end effector.

Manipulator: The robot arm (not including the End of Arm Tooling).

TCP: Tool Center Point. This is the point (coordinate) that we program in relation to.

09: Glossary

Positioning Axes: The first three axes of the robot (1, 2, 3). Base / Shoulder / Elbow = Positioning Axes. These are the axes near the base of the robot.

Orientation Axes: The other joints (4, 5, 6). These joints are always rotary. Pitch / Roll / Yaw = Orientation Axes. These are the axes closer to the tool.

External axis: Motion axis which is not part of the manipulator but which is controlled using the robot controller, e.g. KUKA linear unit, turn-tilt table on the KR150

PART 2: FABRICATION

10: Robotic Fabrication Methods

Fabricating with the KUKA KR10-R1100 Agilus Robot

Now that you have a general understanding of the robotic fabrication process, you can now proceed to the following fabrication methods. It is important to understand that no two methods are exactly the same: each procedure is specific to the end effector that is mounted on the robot and the material you are working with. As mentioned in *Part 1: Fundamentals,* the three fabrication methods that are available at the Design + Technology LAB are:

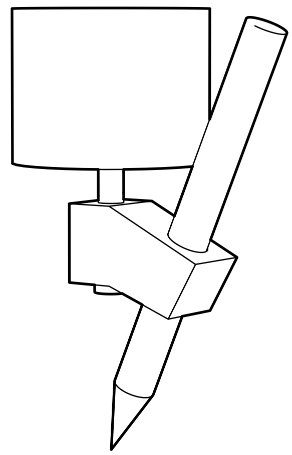
1. **Pen Plotting:** otherwise known as robotic drawing.
2. **3D Printing:** depositing melted plastic layer-by-layer to create a 3d object.
3. **Hot Wire Cutting:** using a hot wire to cut through foam.

11:

Pen Plotting

KUKA KR10-R1100 Designing for Pen Plotting

The following presents instructional information for using the Pen Plotting Tool (shown below) with the KUKA KR10 R1100 at the Design + Technology LAB.



11:

Pen Plotting

STEP 1: Develop your 2D Drawing

STEP 2: Process 2D Vectors

STEP 3: Draw with the Robot

The following section outlines the procedural order of preparing your 2D vector drawing file.

It is your responsibility to prepare a proper file. You will not be able to plot using robot if your file is not prepared adequately.

STEP 1: DEVELOP YOUR 2D DRAWING

2D Digital files can be created using any vector based program of your choice. Common 2D modeling software include Adobe Illustrator, Rhino, Sketchup, AutoCAD,etc.

11: Pen Plotting

STEP 1: Develop your 2D Drawing

STEP 2: Process 2D Vectors

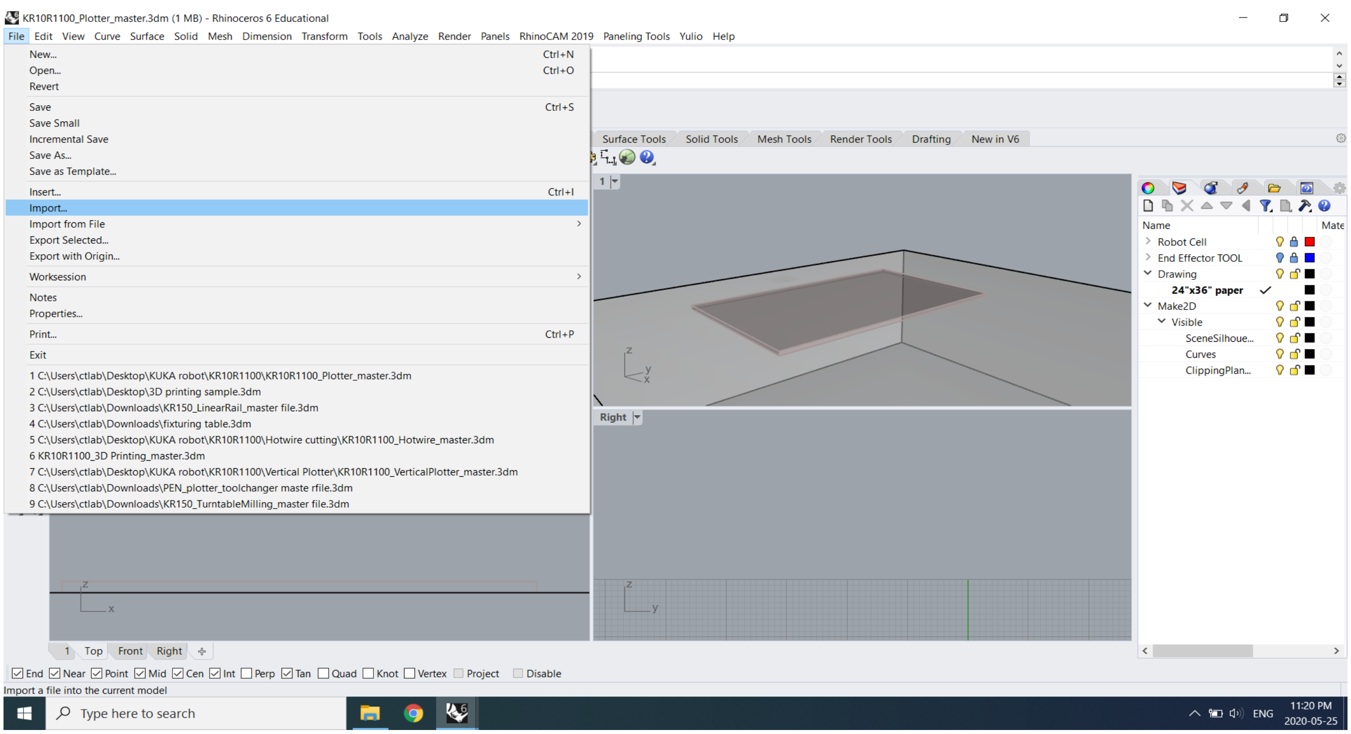
STEP 3: Draw with the Robot

STEP 2: PROCESS 2D VECTORS using RHINO GRASSHOPPER. *(KR10R1100_Plotter_master.gh)*

File preparation

- Ensure file fits on the drawing surface
- Avoid overlapping lines
- Line weights are not necessary

You will be assigned a rhino file with required data: **KR10R1100_Plotter_master.3dm**. *Open the rhino file above and Import your file* ***File > Import*** *select file hit* ***Import.*** Once successfully imported, your model should appear.



11:

Pen Plotting

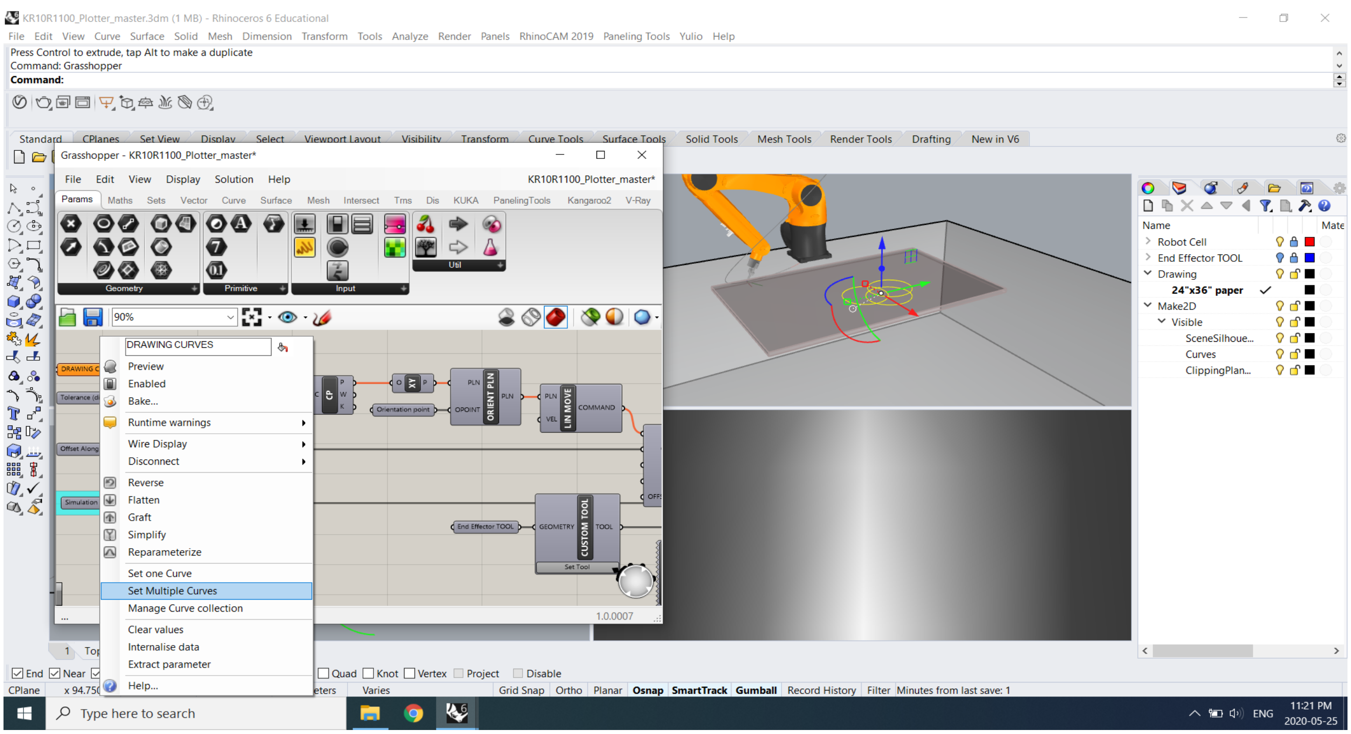
STEP 1: Develop your 2D Drawing

STEP 2: Process 2D Vectors

STEP 3: Draw with the Robot

Grasshopper definition

- Open grasshopper in rhino
- Open allocated 3D printing file in grasshopper:
- KR10R1100_Plotter_master.gh
- In the model space, place the object on top of the MDF spoil board.
- Select and connect all the lines to the "DRAWING CURVES" plugin.
- Always preview using the simulation slider to check for collisions.



Select all the lines and apply to DRAWING CURVES then set as multiple curves.

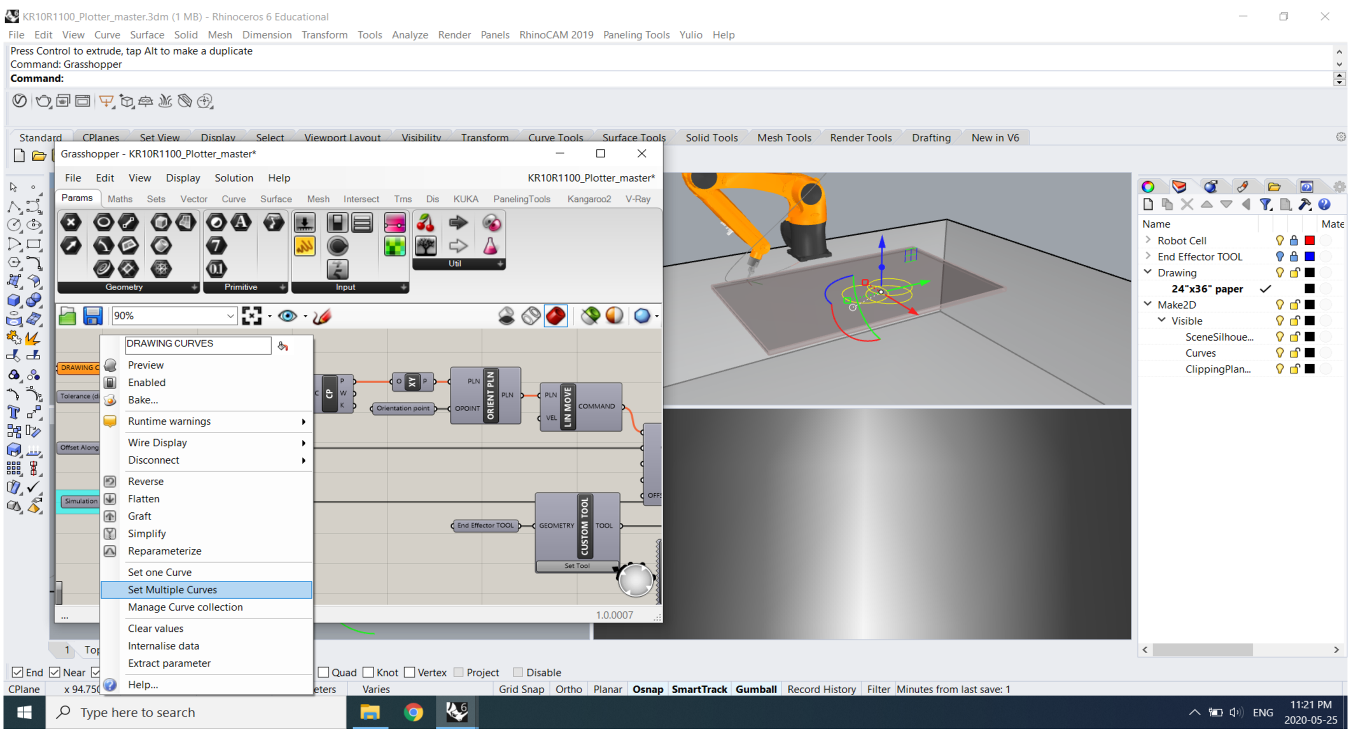
11:

Pen Plotting

STEP 1: Develop your 2D Drawing

STEP 2: Process 2D Vectors

STEP 3: Draw with the Robot



All of the components should be grey before procedure. If not, move to the object to a place robot can reach and check for collisions.

11:

Pen Plotting

STEP 1: Develop your 2D Drawing

STEP 2: Process 2D Vectors

STEP 3: Draw with the Robot

STEP 3: DRAW WITH THE ROBOT

Calibrate tool

Place a pen, pencil, or marker of your choice into the center of the pen tool attachment.

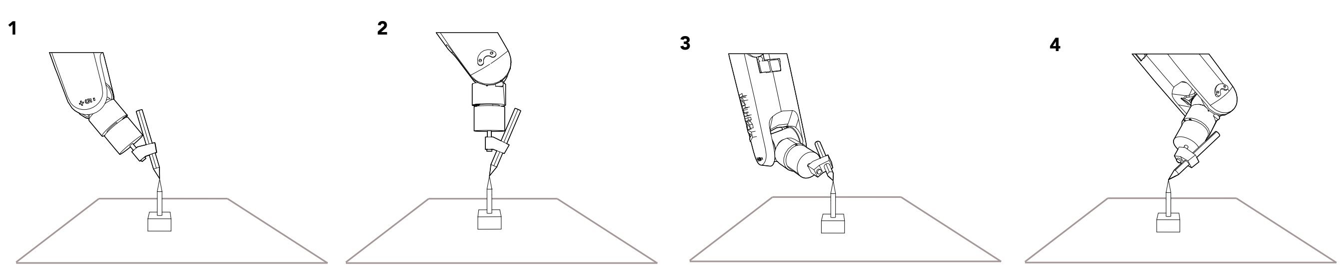
In the teach pendent menu select user group Expert and Operating mode T1 or T2

1. Select *Start-up > Calibrate > Tool >XYZ 4 points.*
2. Assign a number and a name for the tool to be calibrated. Confirm with OK.
3. Move the TCP to a reference point. Confirm with OK.

Move the TCP to the reference point from a different direction 4x. Confirm with OK. Always complete XYZ 4 - Point in World mode.

- X, Y, Z = movement around these axes
- A = Rotation around the X-axis
- B = Rotation around the Y-axis
- C = Rotation around the Z-axis

4. You will be asked to enter the weight of the tool in Kg. For this specific tool enter 2kg.



11:

Pen Plotting

STEP 1: Develop your 2D Drawing

STEP 2: Process 2D Vectors

STEP 3: Draw with the Robot

5. Press Save.

Upon complete Open up KUKA PRC grasshopper definition and enter the previous XYZ 4 point values under the end effector Tool component. Make sure to copy the exact same values from the previous step.

Note: The average error of your the X, Y and Z 4 point should be 0.5 or below to have an accurate calibrated tool.

Calibrating surface

Your drawing surface may vary; As a result you need to figure out the location of your surface correlating with the rhino model.

“KR10R1100_Plotter_master.3dm” has the base location to start your drawing.

Operate in T1 mode in SmartPad to simulate the drawing path and check for possible collisions. If there are possible collisions, release the Play button to stop the simulation.

Go back to your initial Rhino drawing and adjust and repeat.

12:

3D Printing

KUKA KR10-R1100 3D Printing

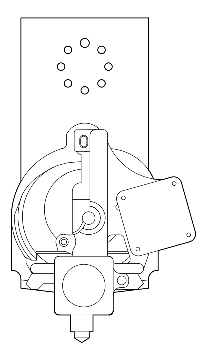
The following presents instructional information on 3D Printing with the KUKA KR10 R1100 at the Design + Technology LAB.

FDM 3D Printing

FDM (Fused Deposition Modeling): An additive manufacturing process where a thermoplastic filament (spool or pellets) is pushed through a heated nozzle, which selectively deposits the molten material onto a platform, where it cools and solidifies.

FDM printing extruder is attached to the end of the KUKA robotic arm that feeds spools or pellets of thermoplastic filament through a hot-end and extrudes molten material through its nozzle, building vertical layers. These stacked layers cool and harden, resulting in a solid 3D printed object. It is similar to conventional 3D printing extruders; however, it will print continuously and can print at larger scales.

Caution: FDM 3D printer hot-ends, and nozzles can reach temperatures of 300 degrees, and cause severe burns. Do not touch these components during preheating, printing or immediately post-printing - for a minimum of 30 minutes after



12:

3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot

KUKA Robotic 3D Printing General Notes

- NO overhangs beyond 45 degrees. Do not exceed more than a 45 degree angle on any given surface of your model. Exceeding 45 degrees means every new layer has less of the previous layer to bond with, resulting print failure.
- Since it is a continuous printing process ONLY PRINT ONE OBJECT AT A TIME or you must consider/design the extrusion path between objects. .
- Always print in section if printing large object

The following section outlines the procedural order for preparing your 3D Printing 3D Model file:

STEP 1: DEVELOP YOUR 3D MODEL FILES

3D model can be created using any software of your choice. Common 3D modeling software include Rhino, Fusion 360, Sketchup, AutoCAD, SolidWorks, 3ds Max etc.

12:
3D Printing

STEP 2: SLICE YOUR MODEL

Slice your model using the RHINO GRASSHOPPER Master File. **(KR10R1100_3D Printing_master.gh)**

Boolean Union individual components

- After you have completed your 3D model, carefully inspect it for intersecting and/or open geometries.
- If your model is a collection of individual forms, these will need to be joined into one form. To do so, select the components to be 3D printed and type the Rhino command "Boolean Union".
- If there are no errors with your model components, this command should join all your individually modeled components into one geometry.
- To confirm the success of the command, click your model and see whether doing so selects your entire model for 3D printing.
- Use the command CONTOUR to set layer heights. Height determines the thickness of each layer; thus, affecting the resolution of the printed model. You are able to select any value between 0.7mm-1.2mm. A higher layer height setting would result in a faster, less refined print, and is recommended for first prototypes. *Please look at individual setting for types of 2D printing extruders.*

It is your responsibility to prepare a proper model file. You will not be able to 3D print if your model is not prepared adequately.

12:

3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

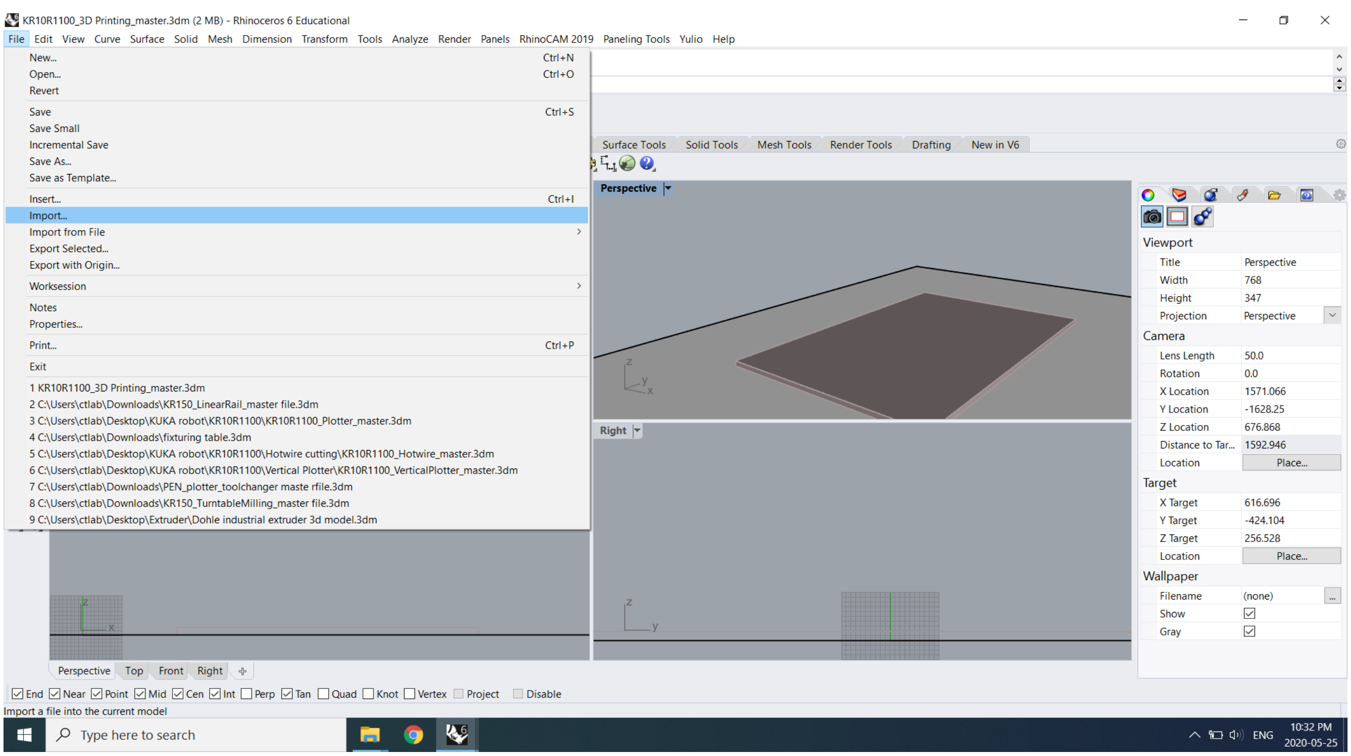
STEP 3: Print with the Robot

If you are using other software to create the 3D object, simply import the file into rhino and follow the procedure below.

You will be assigned a rhino file with required data: in this case: **KR10R1100_3D Printing_master.3dm**. Open the rhino file above and Import your file:

File > Import select file hit Import.

Once successfully imported, your model should appear on the rhino.



12:

3D Printing

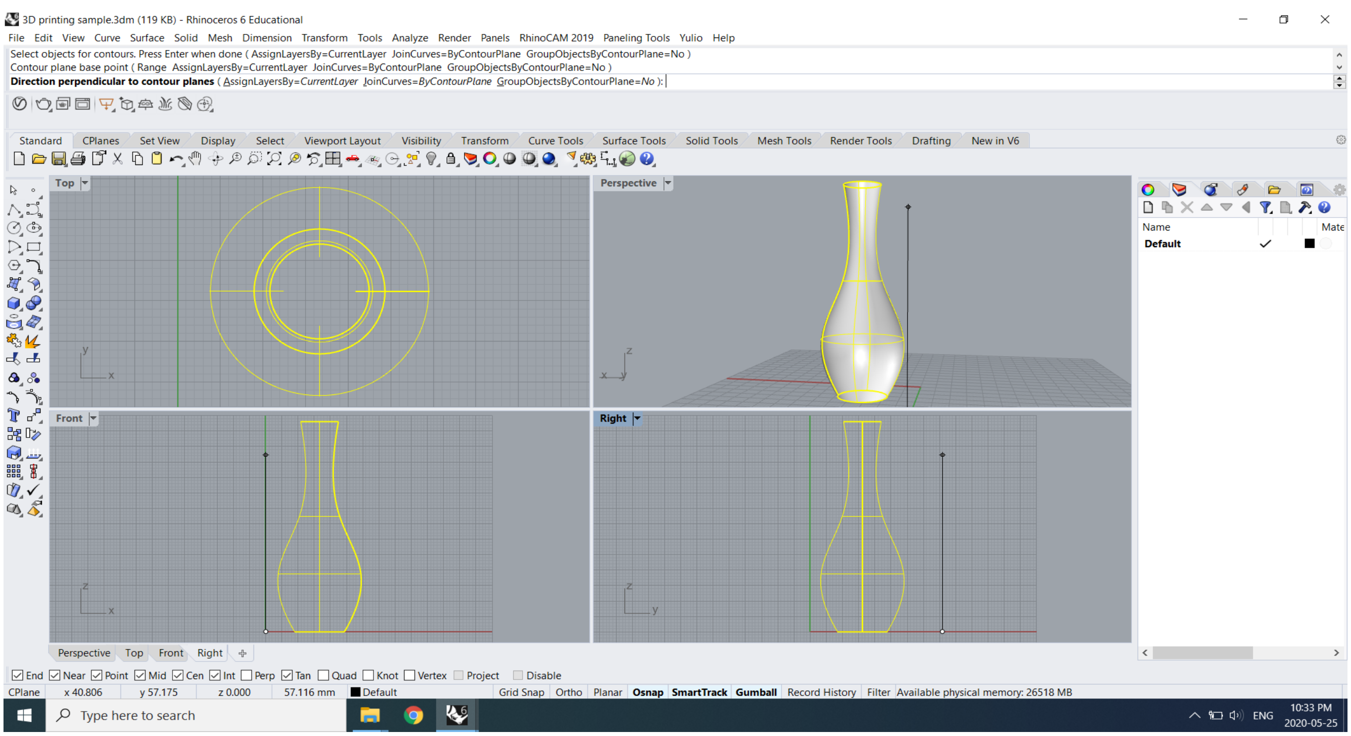
STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot

Open grasshopper in rhino

- Open allocated 3D printing file in grasshopper: KR10R1100_3D Printing_master.gh
- Place the object on top of the 3D printing platform or known/teachable reference surface.
- Select and connect all the contour lines to the "3D OBJECT" plugin. Always preview using simulation slider to check if printing correctly. If the robot moving in the opposite direction use "Reverse" plugin to change the direction.
- Select all the contours and apply to 3D DRAWING CURVES, set as multiple curves (shown below).



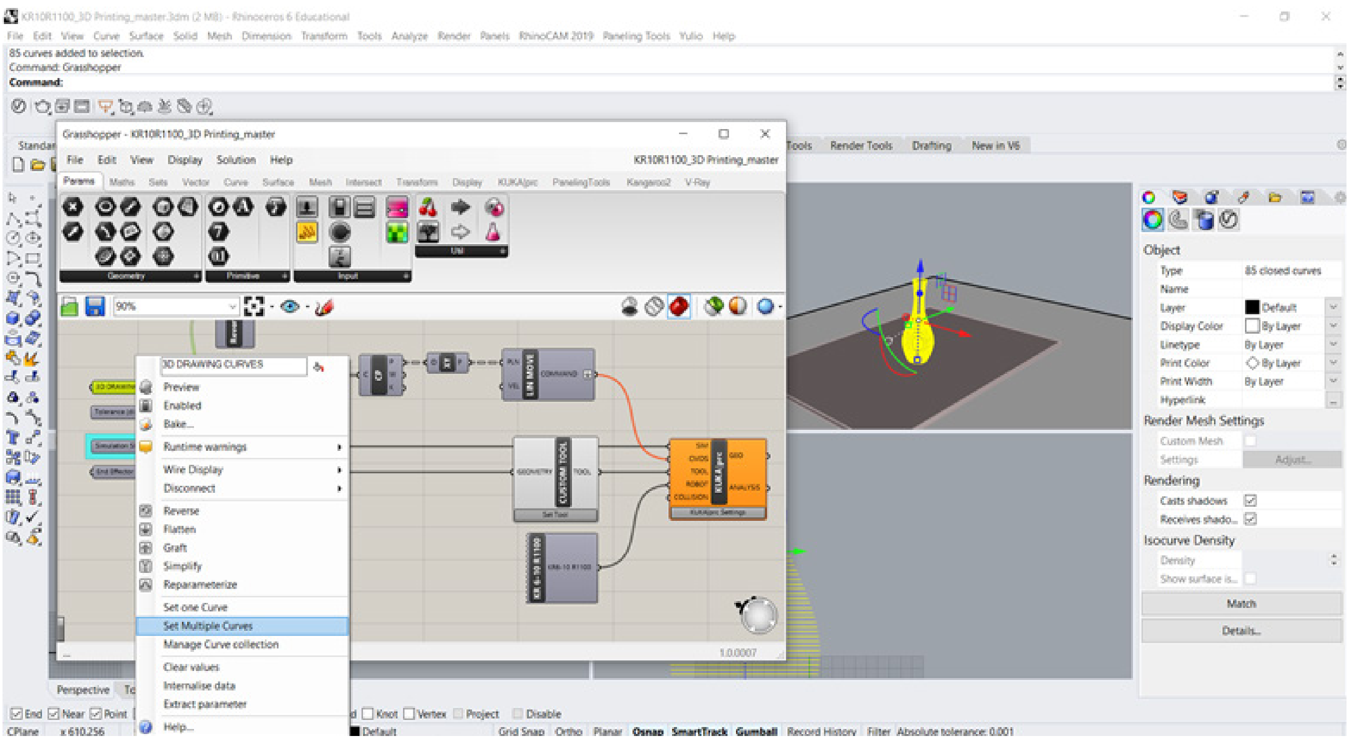
Be sure that the direction of the contour is perpendicular to the ground. Z direction

12:
3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot



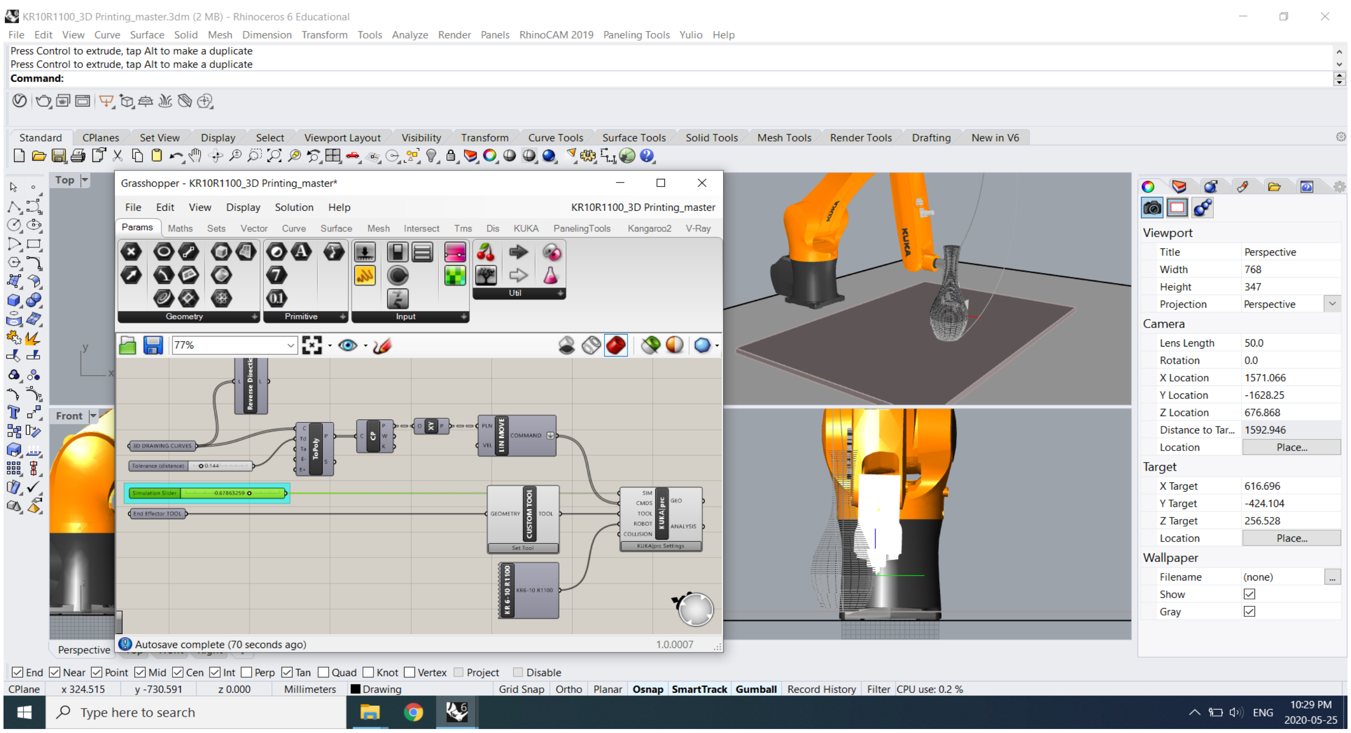
12:

3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot



If the direction of the tool path not correct, apply “Reverse Direction” component to the drawing curves

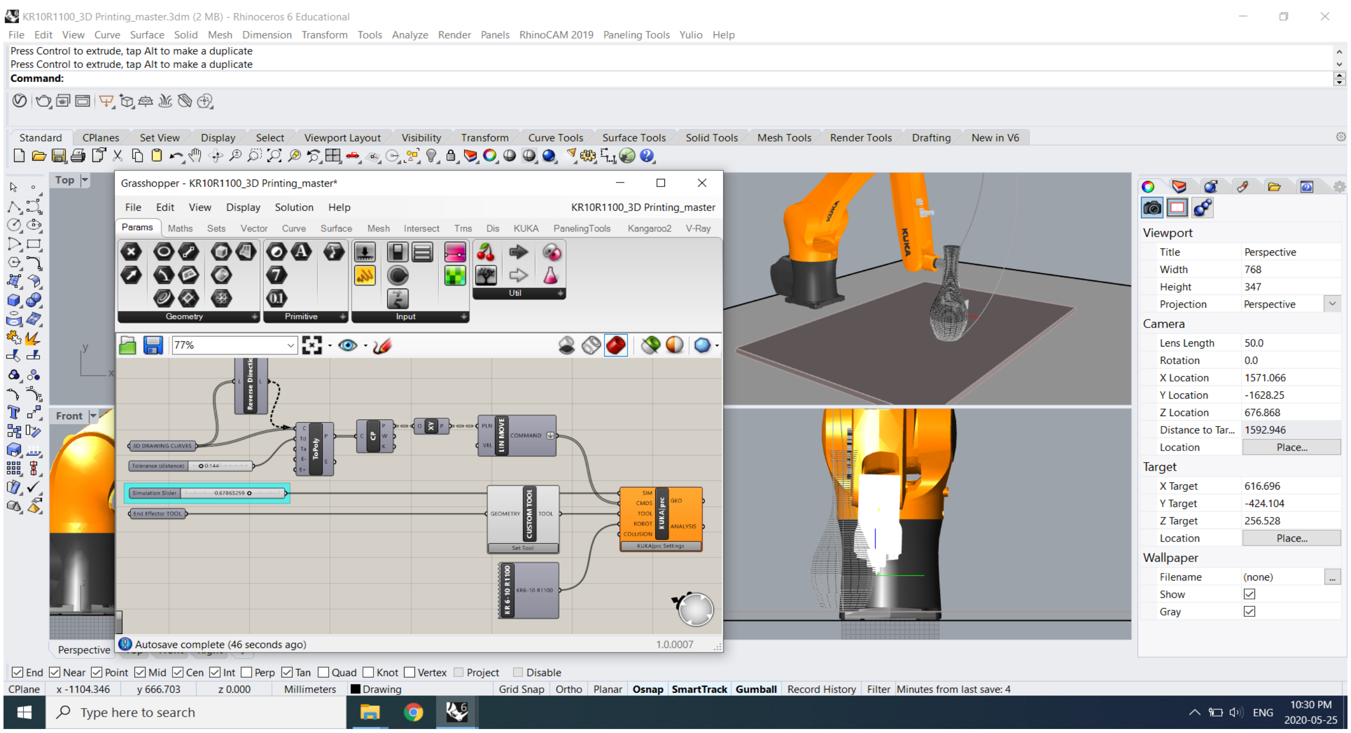
12:

3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot



If the direction of the tool path not correct, apply “Reverse Direction” component to the drawing curves

12:

3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot

STEP 3: PRINT WITH THE ROBOT

On your appointment the 3D printing extruder will be attached to the robot arm, if not ask a technician for help.

Calibrate Tool

In the teach pendant menu select user group Expert and Operating mode T1 or T2

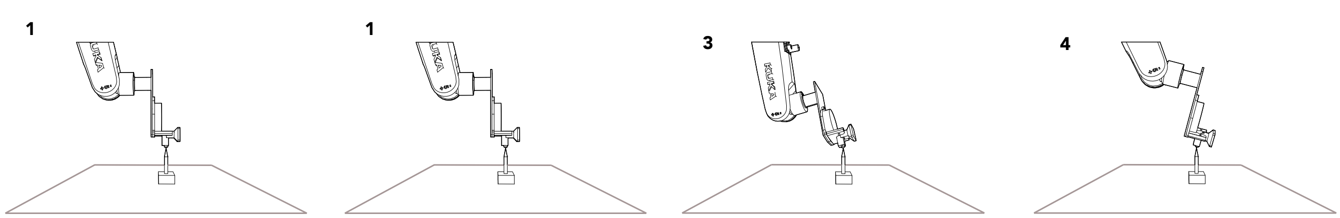
1. Select Start-up > Calibrate > Tool >XYZ 4 points.
2. Assign a number and a name for the tool to be calibrated. Confirm with OK.
3. Move the TCP to a reference point. Confirm with OK.
4. Move the TCP to the reference point from a different direction 4x. Confirm with OK. Always complete XYZ 4 - Point in World mode.

1) X, Y, Z = movement around these axes

2) A = Rotation around the X-axis

3) B = Rotation around the Y-axis

4) C = Rotation around the Z-axis



12:

3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot

5. You will be asked to enter the weight of the tool in Kg. For this specific tool enter **2kg**.
6. Press Save.

Upon complete Open up KUKA PRC grasshopper definition and enter the previous XYZ 4 point values under the **end effector Tool** component. Make sure to copy the exact same values from the previous step.

Note: The average error of your X, Y and Z 4 point should be 0.5 or below to have an accurate calibrated tool.

3D printer

- Connect the USB cable to the Design + Technology LAB laptop
- Turn on the 3D printer attached to the robot arm, you can find the power supply unit attached to the robot.

Simplify 3D

- Open simplify 3D software on the laptop.
- Got to Manage Control Panel (tool panel on the right side of the interface)
  - Hit “Connect”
  - Slide the Fan Speed slider up all the way
  - Check whether the both fans are on (top + nozzle fan)
  - Slide the Fan Speed slider down to 50%
  - Hit ON for the extruder (once actually on, should say Set)
  - In the top tabs of Simplify 3D, go to “Jog …”

12:

3D Printing

STEP 1: Develop your 3D Model

STEP 2: Slice your Model

STEP 3: Print with the Robot

Open “Command Prompt” on the laptop (window search on computer)

- Type “cd Desktop” and hit Enter
- Type “python clicker.py”
- Hover the mouse over the 100mm extrude button and hit Enter to activate the command in Command Prompt program.
- This will begin the filament extrusion

Kuka Robot

- Simulate your program in T1 mode to check if the robot is printing properly.
- Hold up the extruded filament as the robot moves towards the print surface to ensure it doesn’t get in the way of the first layer. Make sure the first layer adheres properly to the bed surface.

Calibrate surface

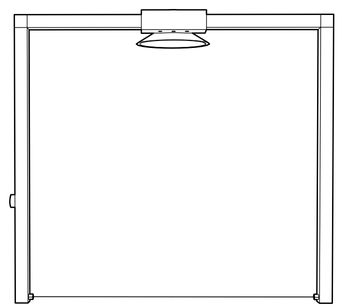
- Your printing platform may vary: As a result you will need to figure out the perfect height of printing platform for layer to adheres properly
- “KR10R1100_3D Printing_master.3dm” has the base location to start your drawing.
- Operate in T1 mode in SmartPad to simulate the drawing path and check for possible collisions. If there are possible collisions, release the Play button to stop the simulation.
- Go back to your initial Rhino drawing and adjust and repeat.

13:

Hot Wire Cutting

KUKA KR10-R1100 Hot Wire Cutting

The following manual presents instructional information on Hot Wire Cutting Tool (shown below) using KUKA KR10 R1100 at the Design + Technology LAB.



13:
Hot Wire Cutting

STEP 1: Develop your 3D Model

STEP 2: Process your 3D Model

STEP 3: Print with the Robot

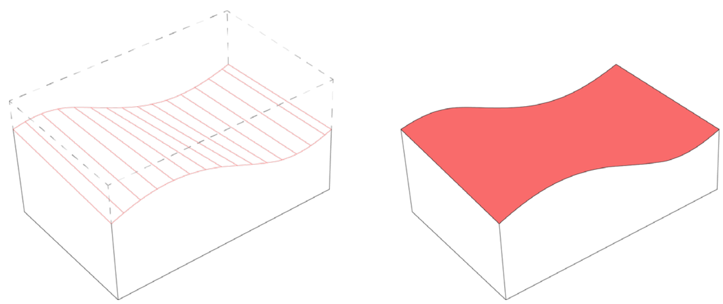
The following section outlines the procedural order of preparing your Hot Wire cutting file.

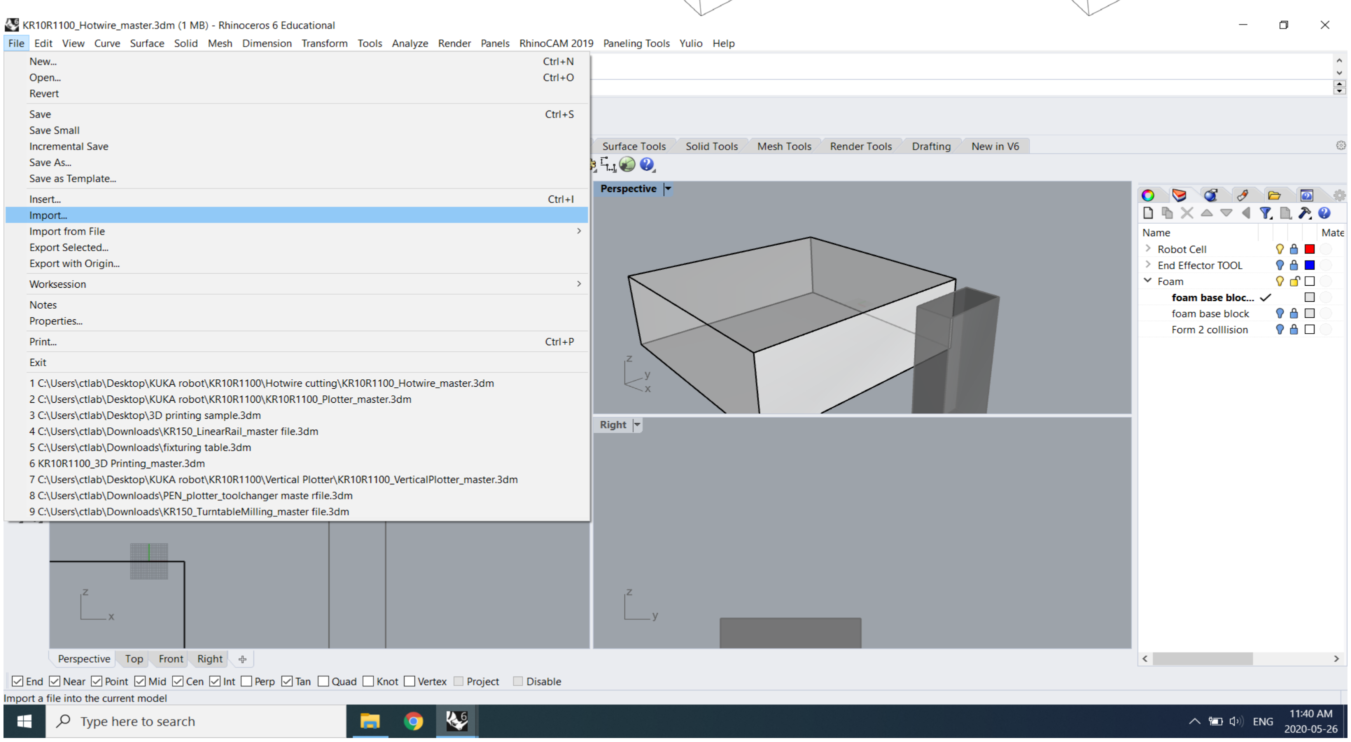
STEP 1: DEVELOP YOUR 3D MODEL FILE

3D model files can be created using any software of your choice. Common 3D modeling software include Rhino, Fusion 360, Sketchup, AutoCAD, SolidWorks, 3ds Max etc.

All cutting planes require to be in x, y direction

Model should be created using 2 curves, simply loft together. The lines in-between are the areas where the hot wire will pass through.





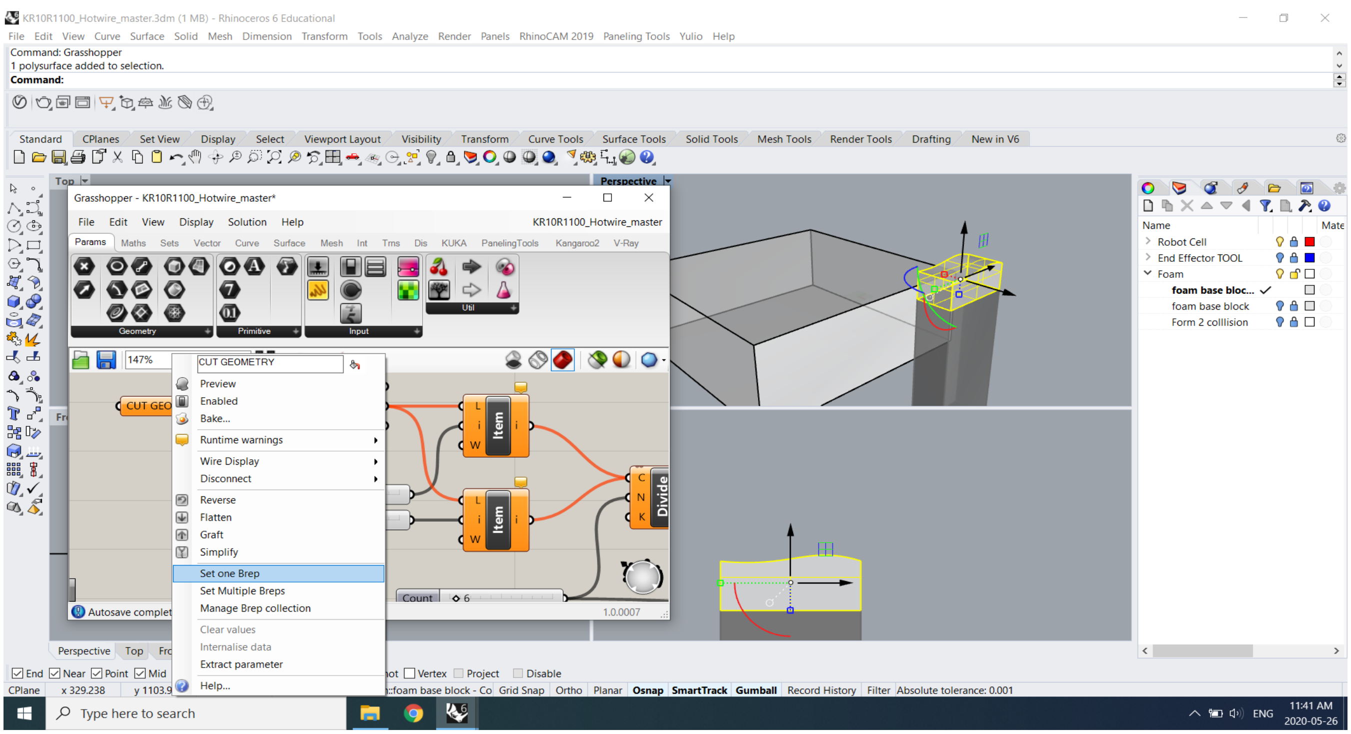
Import a model into Rhino.

13:

Hot Wire Cutting

STEP 2: PROCESS YOUR MODEL through RHINO GRASSHOPPER. (KR10R1100_Hotwire_master.gh)

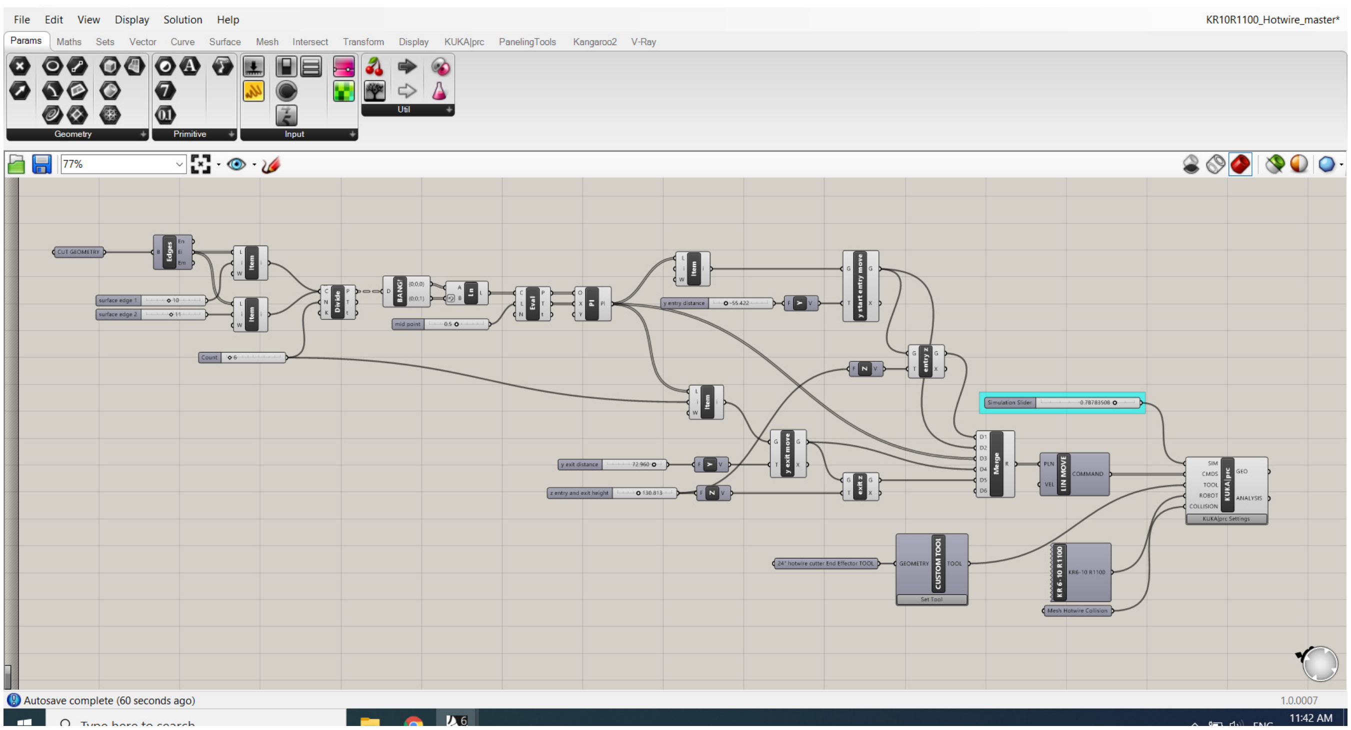
- Open grasshopper in rhino
- Open allocated 3D printing file in grasshopper: **KR10R1100_Hotwire_master.gh**
- In model space place the object on top of the 3D printing plinth.
- Select the object and connect to CUT GEOMETRY plugin. Always preview using simulation slider to check for collision.



Select the geometry and apply to CUT GEOMETRY then adjust surface edges for accuracy.

13:

Hot Wire Cutting



All of the components should be grey before procedure, if not move to the object to a place robot can reach and check for collision.

13:
Hot Wire Cutting

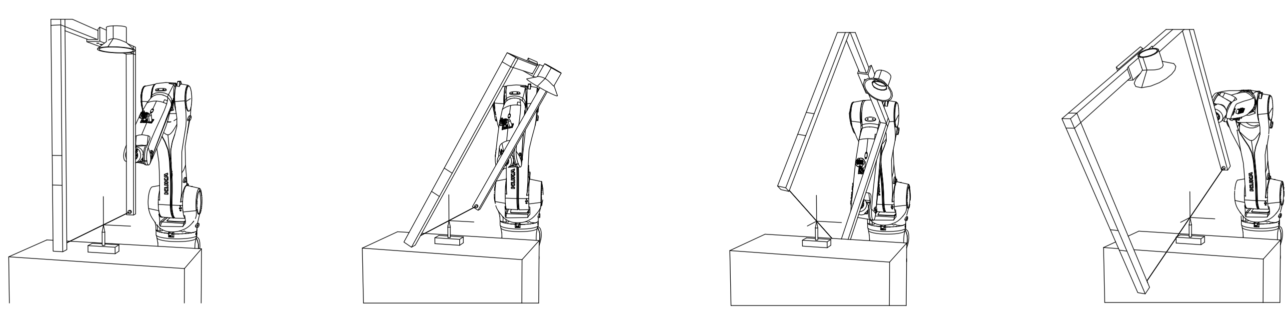
STEP 3: PRINT WITH THE ROBOT

On your appointment the hot Wire cutter will be attached to the robot arm, if not ask a technician for help.

Calibrating tool

1. In the teach pendent menu select user group Expert and Operating mode T1 or T2
2. Select *Start-up > Calibrate > Tool >XYZ 4 points.*
3. Assign a number and a name for the tool to be calibrated. Confirm with OK.
4. Move the TCP to a reference point. Confirm with OK.
5. Move the TCP to the reference point from a different direction 4x. Confirm with OK. Always complete XYZ 4 - Point in World mode.

- X, Y, Z = movement around these axes
- A = Rotation around the X-axis
- B = Rotation around the Y-axis
- C = Rotation around the Z-axis



13:

Hot Wire Cutting

STEP 1: Develop your 3D Model

STEP 2: Process your 3D Model

STEP 3: Print with the Robot

6. You will be asked to enter the weight of the tool in Kg. For this specific tool enter **5kg.**
7. Press Save.

Once complete, open up KUKA PRC grasshopper definition and enter the previous XYZ 4 point values under the end effector Tool component. Make sure to copy the exact same values from the previous step.

Note: The average error of your the X, Y and Z 4 point should be 0.5 or below to have an accurate calibrated tool.

Magnum Controller

- Set Magnum Controller setting for wire cutter 53 – 55 degree
- Set the fume vacuum on “high”
- Temperature of the wire cutter drops when the vacuum is turned ON

Kuka Robot

Simulate your program in T1 mode to check if the robot is moving properly.

13:

Hot Wire Cutting

STEP 1: Develop your 3D Model

STEP 2: Process your 3D Model

STEP 3: Print with the Robot

Calibrating surface

- Your printing platform may vary: As a result you will need to figure out the perfect height for robot to perform the task.
- “KR10R1100_Hotwire_master.3dm” has the base location to start your drawing.
- Operate in T1 mode in SmartPad to simulate the drawing path and check for possible collisions. If there are possible collisions, release the Play button to STOP the simulation.
- Go back to your initial Rhino drawing and adjust and repeat.